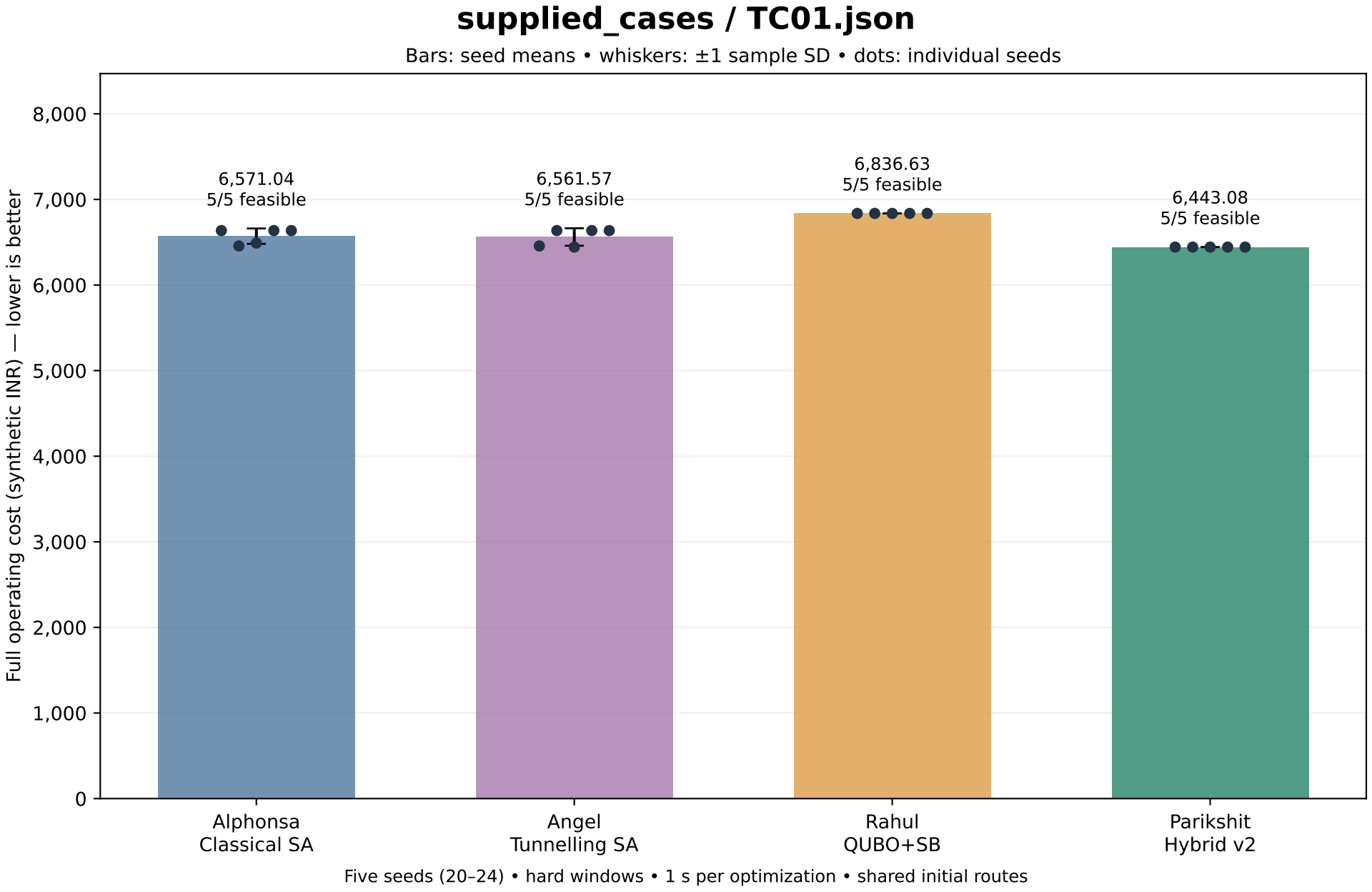
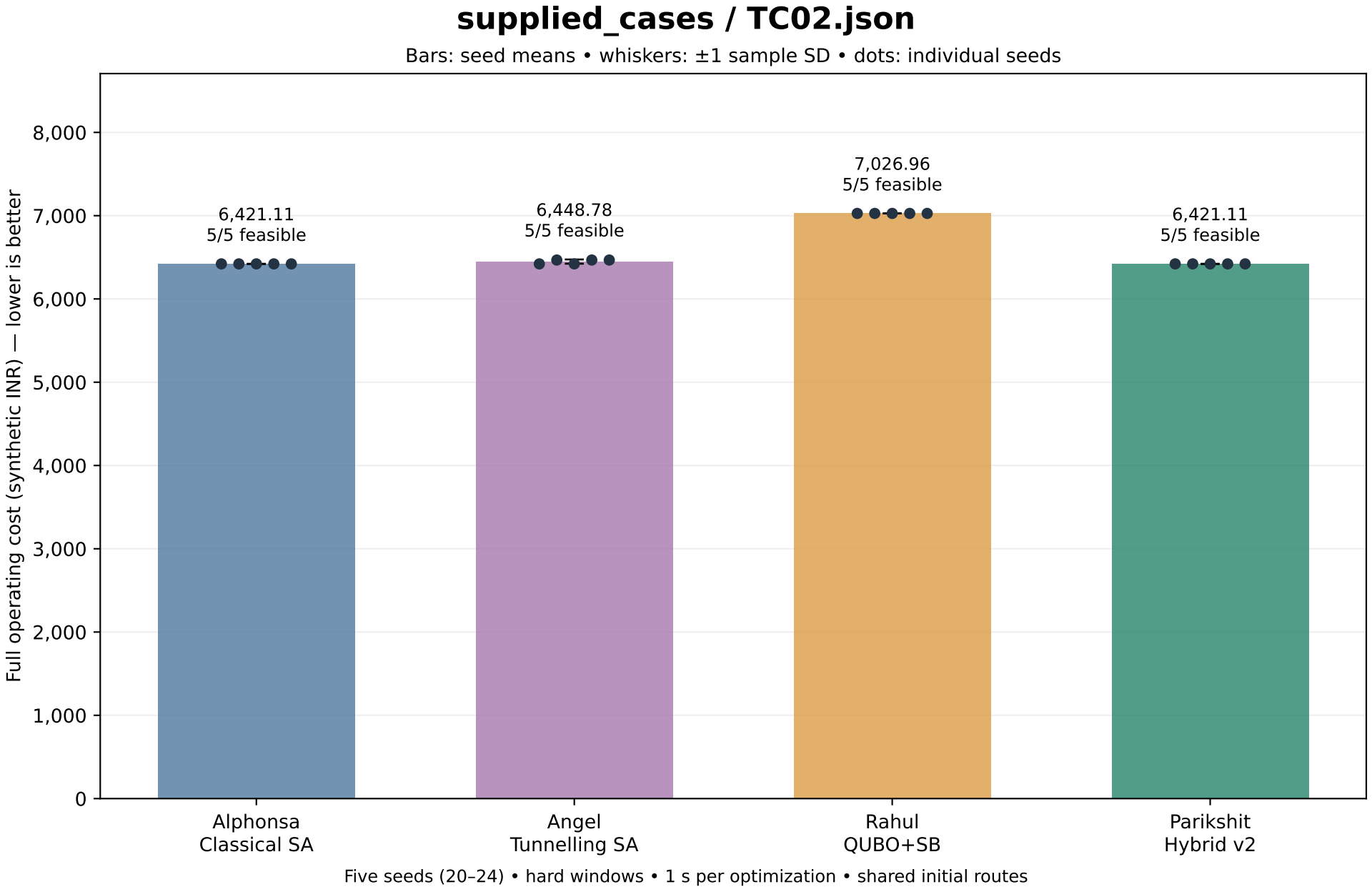
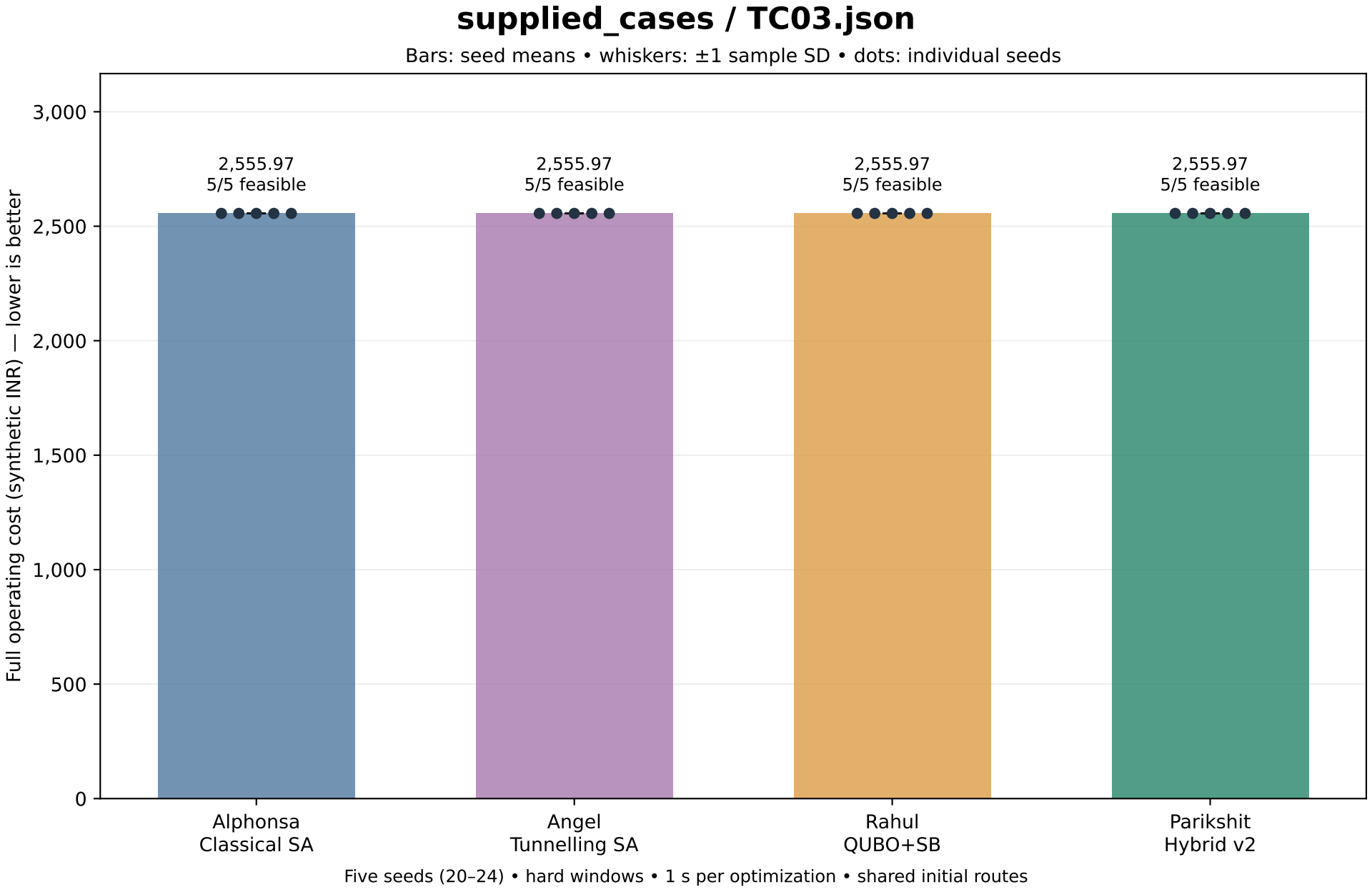
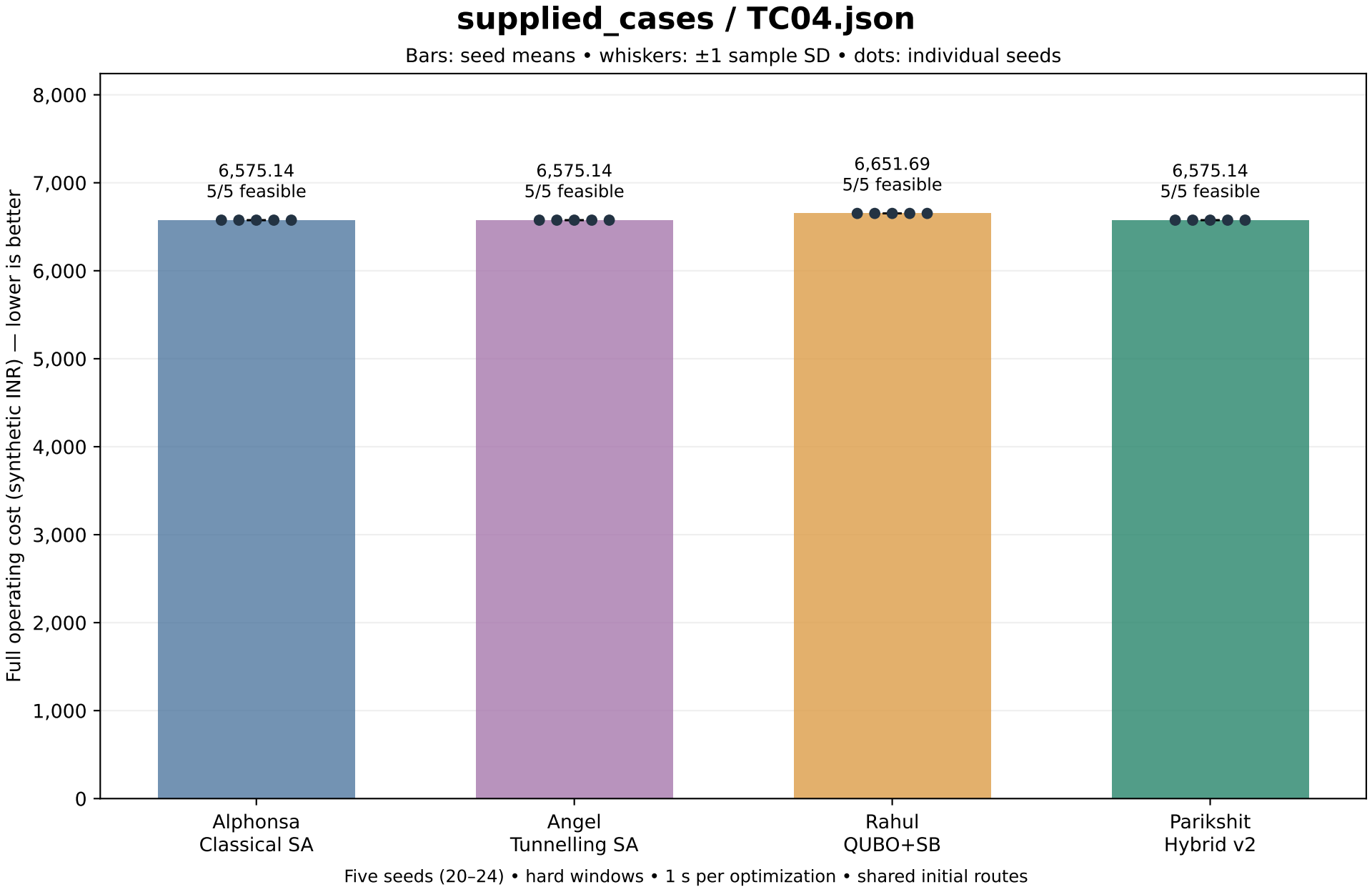
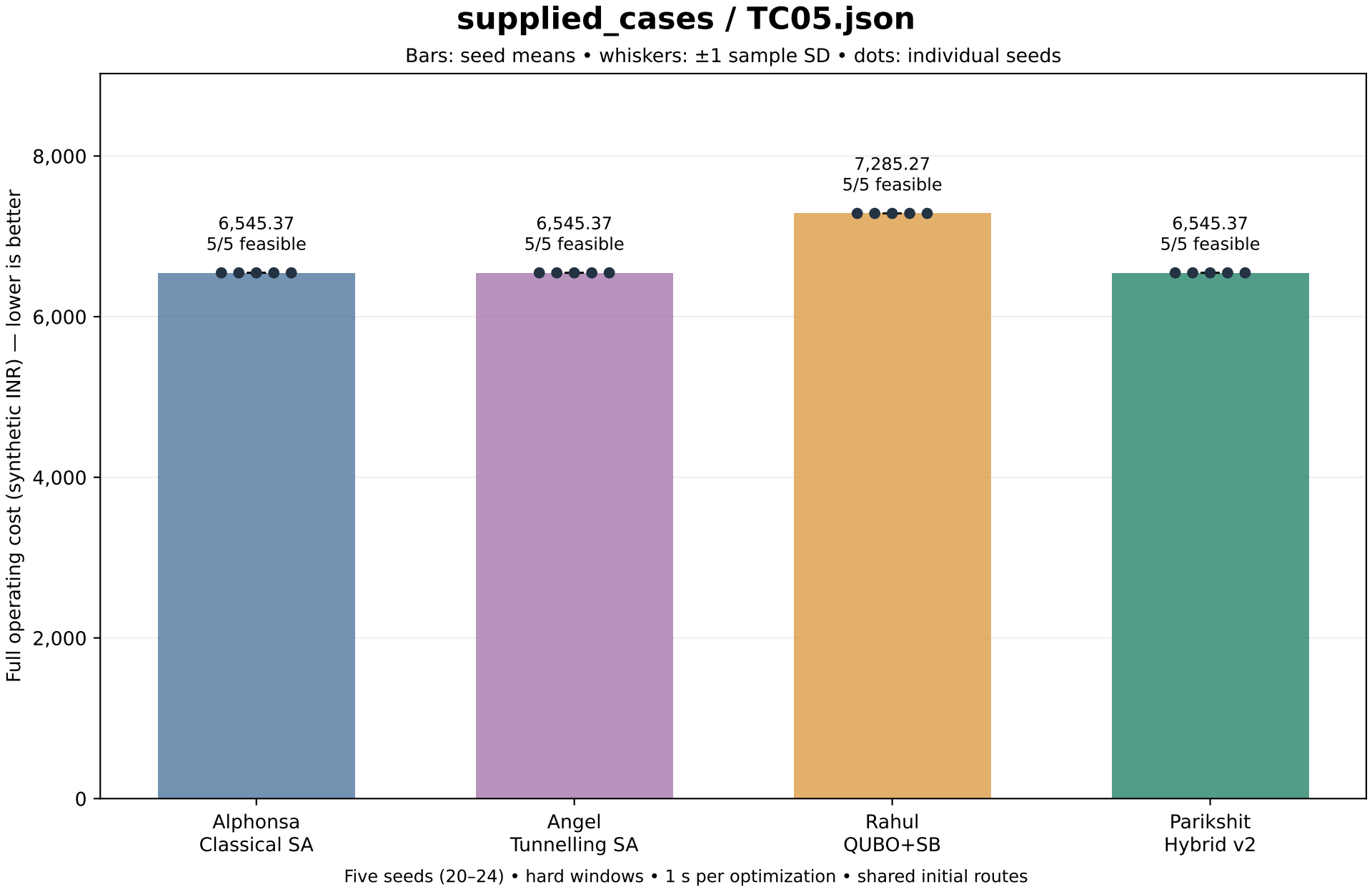
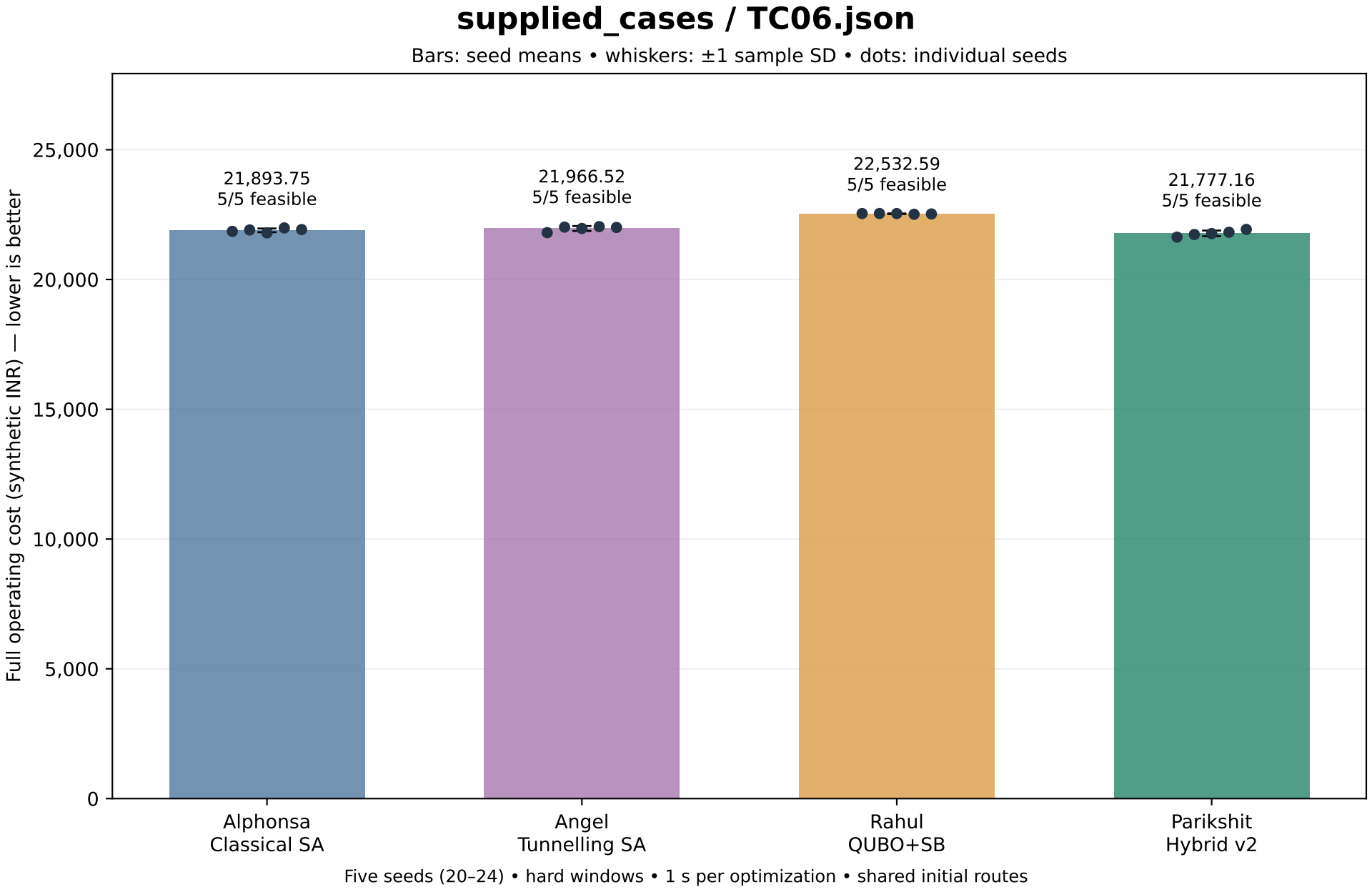

supplied_cases / TC01.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
6,571.04
5/5 feasible
6,561.57
5/5 feasible
6,836.63
5/5 feasible
6,443.08
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

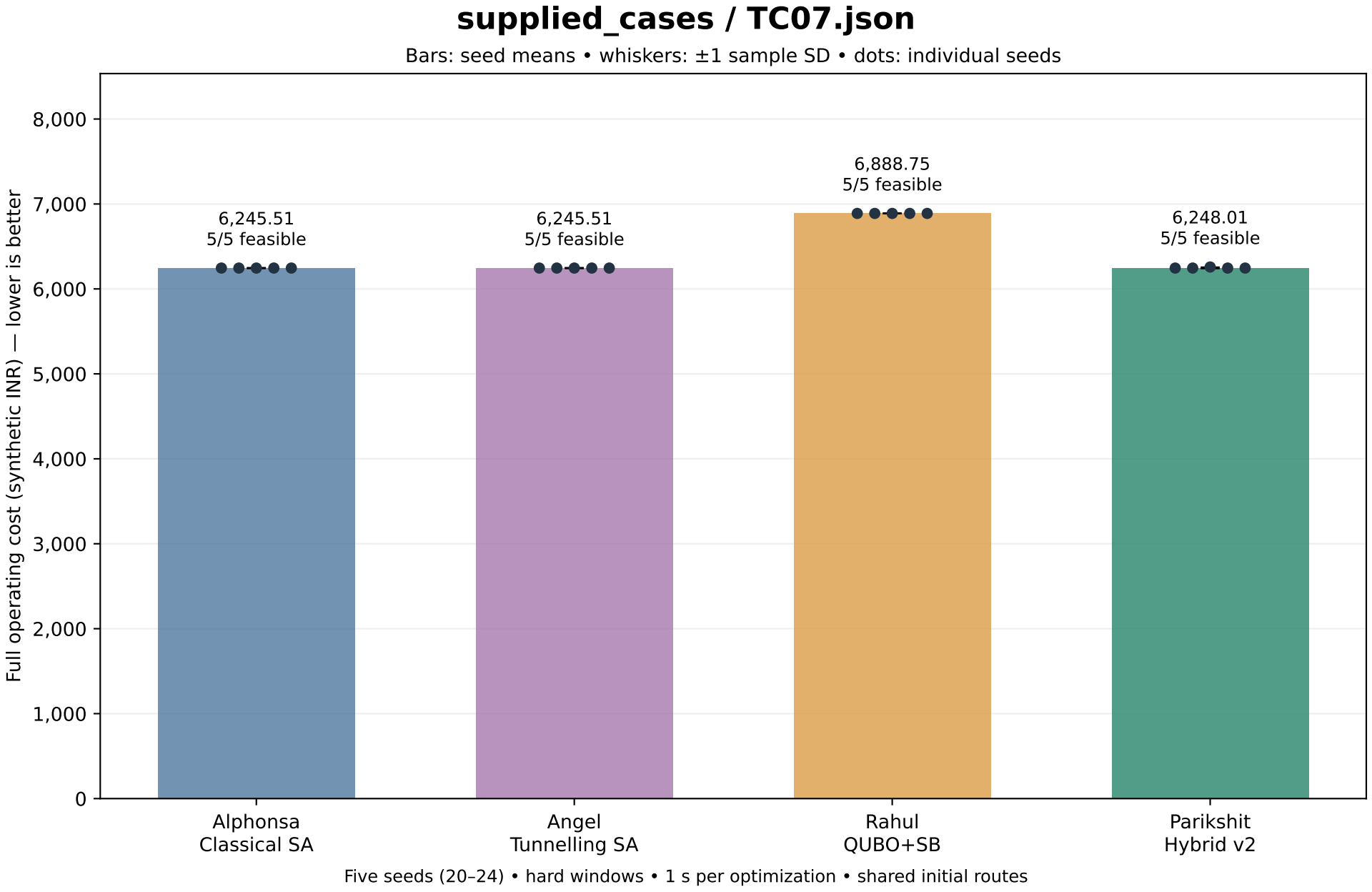

supplied_cases / TC02.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
6,421.11
5/5 feasible
6,448.78
5/5 feasible
7,026.96
5/5 feasible
6,421.11
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

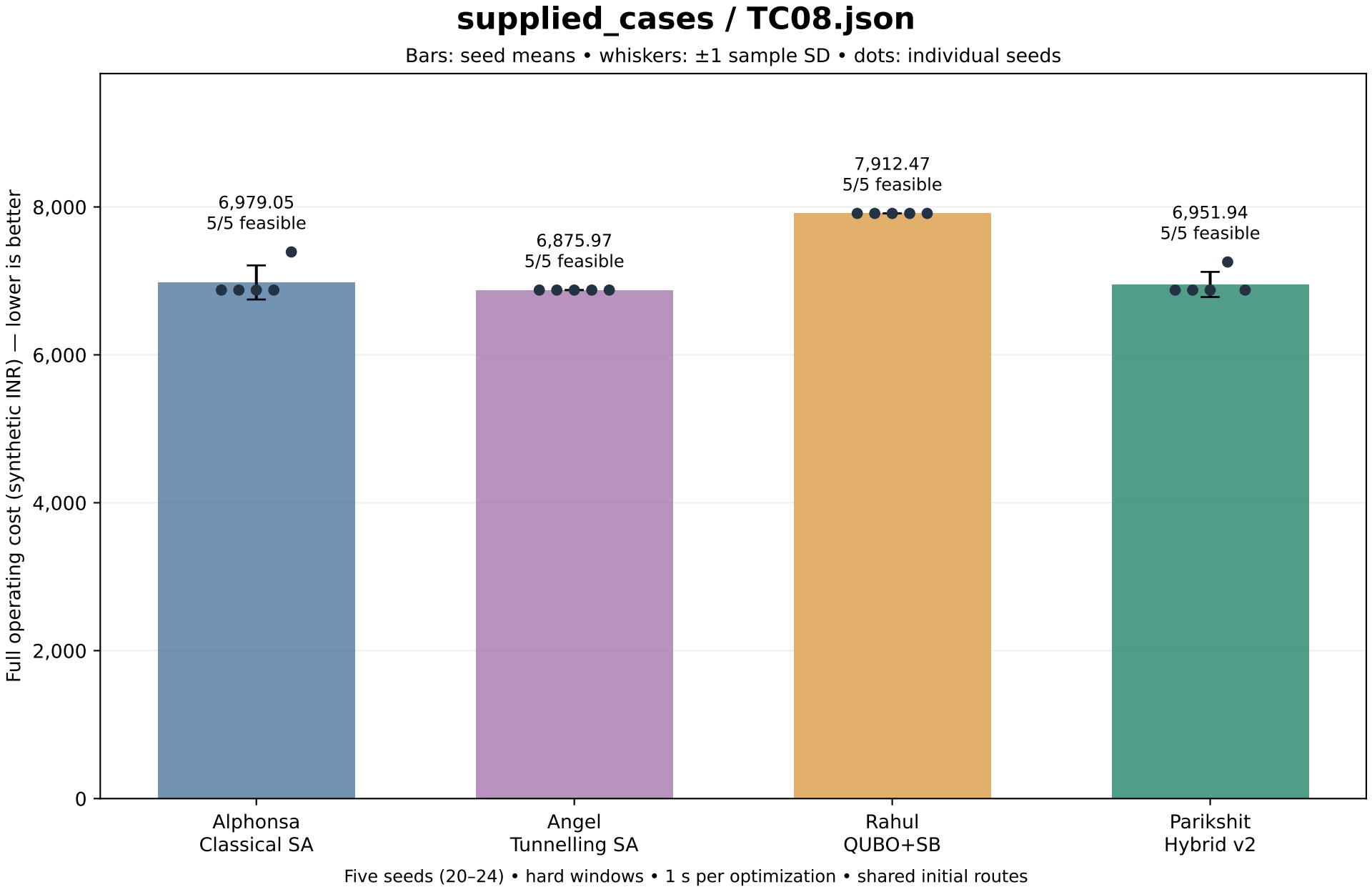

supplied_cases / TC03.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
3,000
2,500
2,000
1,500
1,000
500
0
2,555.97
5/5 feasible
2,555.97
5/5 feasible
2,555.97
5/5 feasible
2,555.97
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

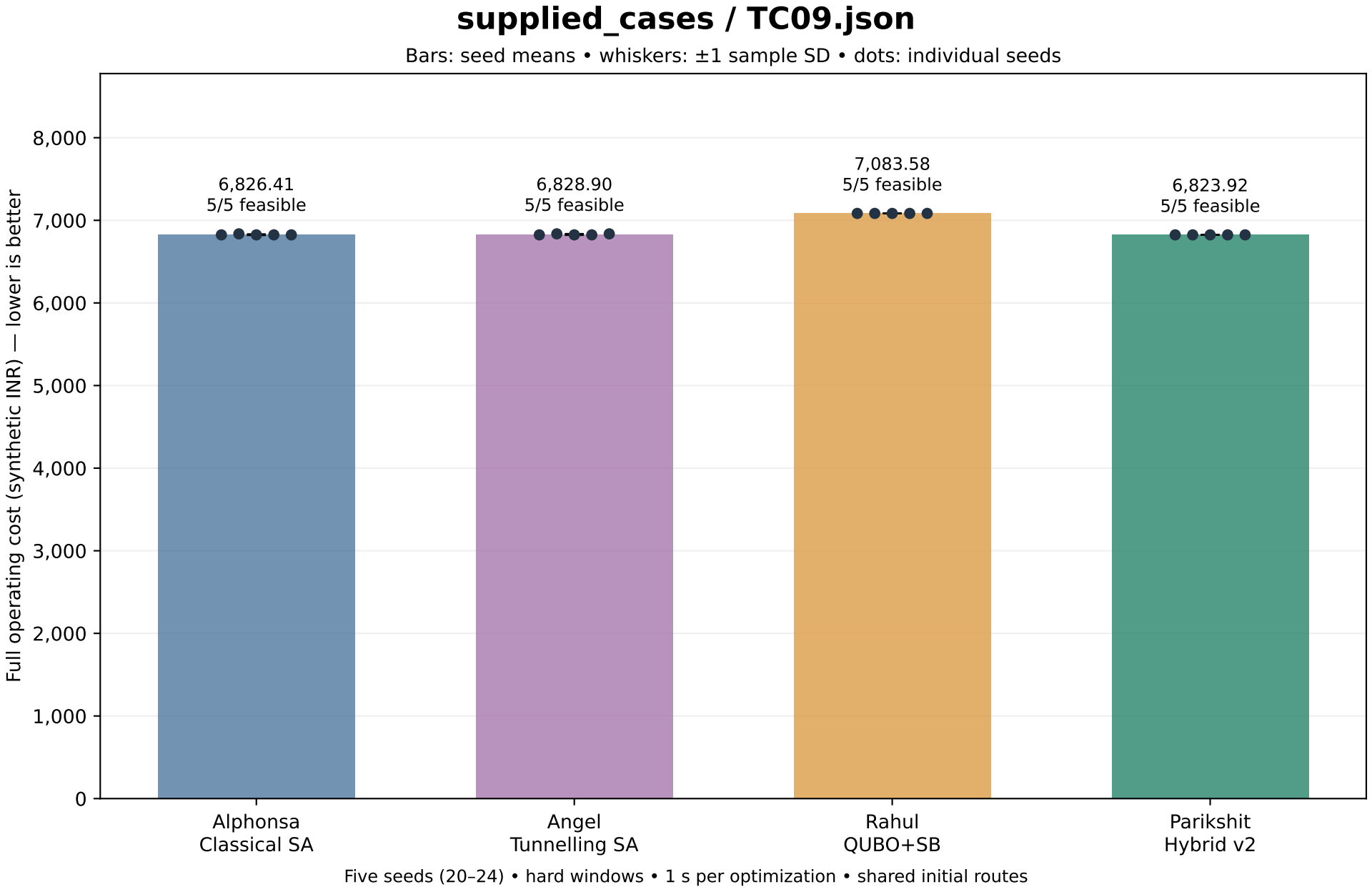

supplied_cases / TC04.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
6,575.14
5/5 feasible
6,575.14
5/5 feasible
6,651.69
5/5 feasible
6,575.14
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

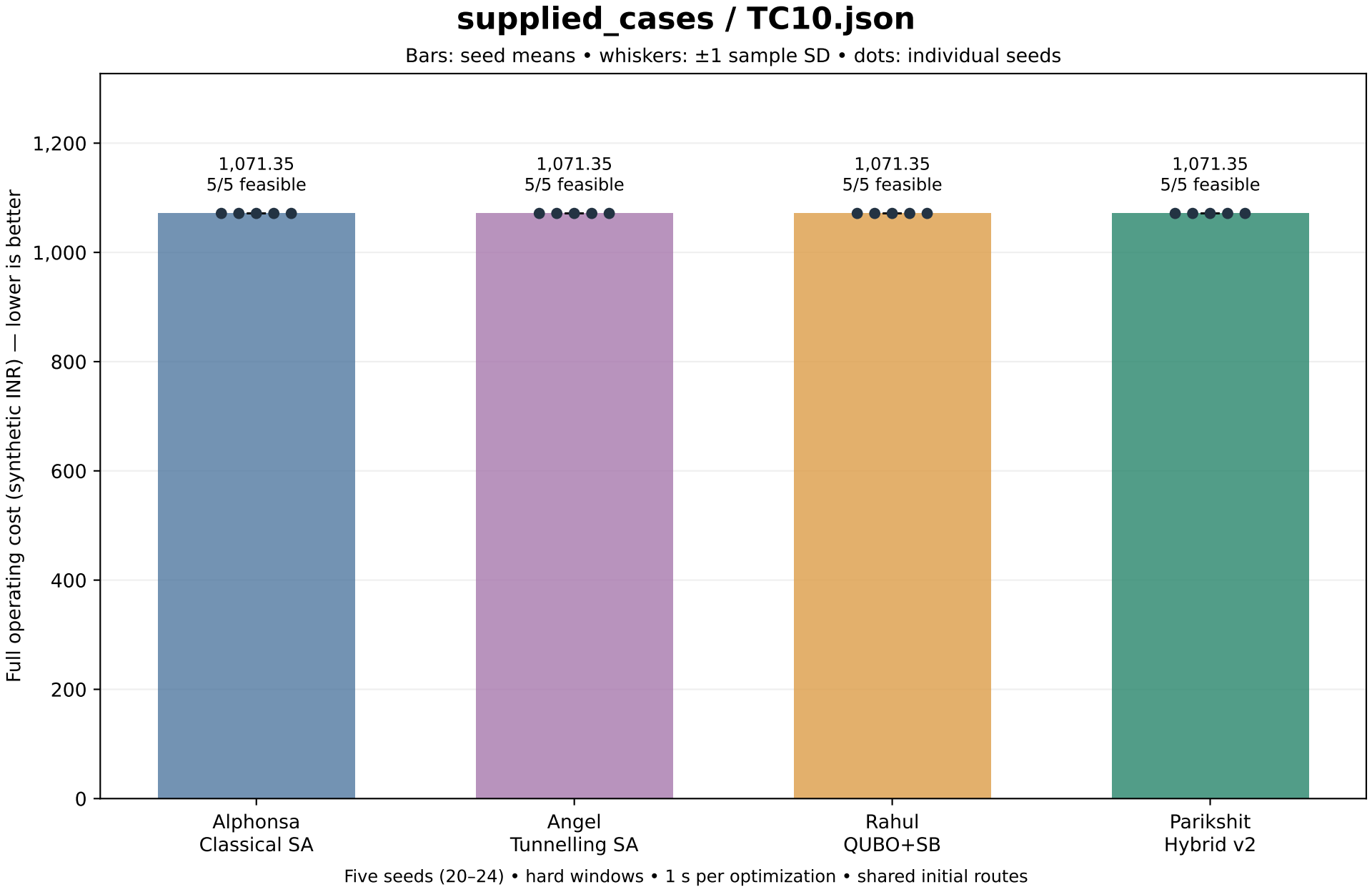

supplied_cases / TC05.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
6,000
4,000
2,000
0
6,545.37
5/5 feasible
6,545.37
5/5 feasible
7,285.27
5/5 feasible
6,545.37
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

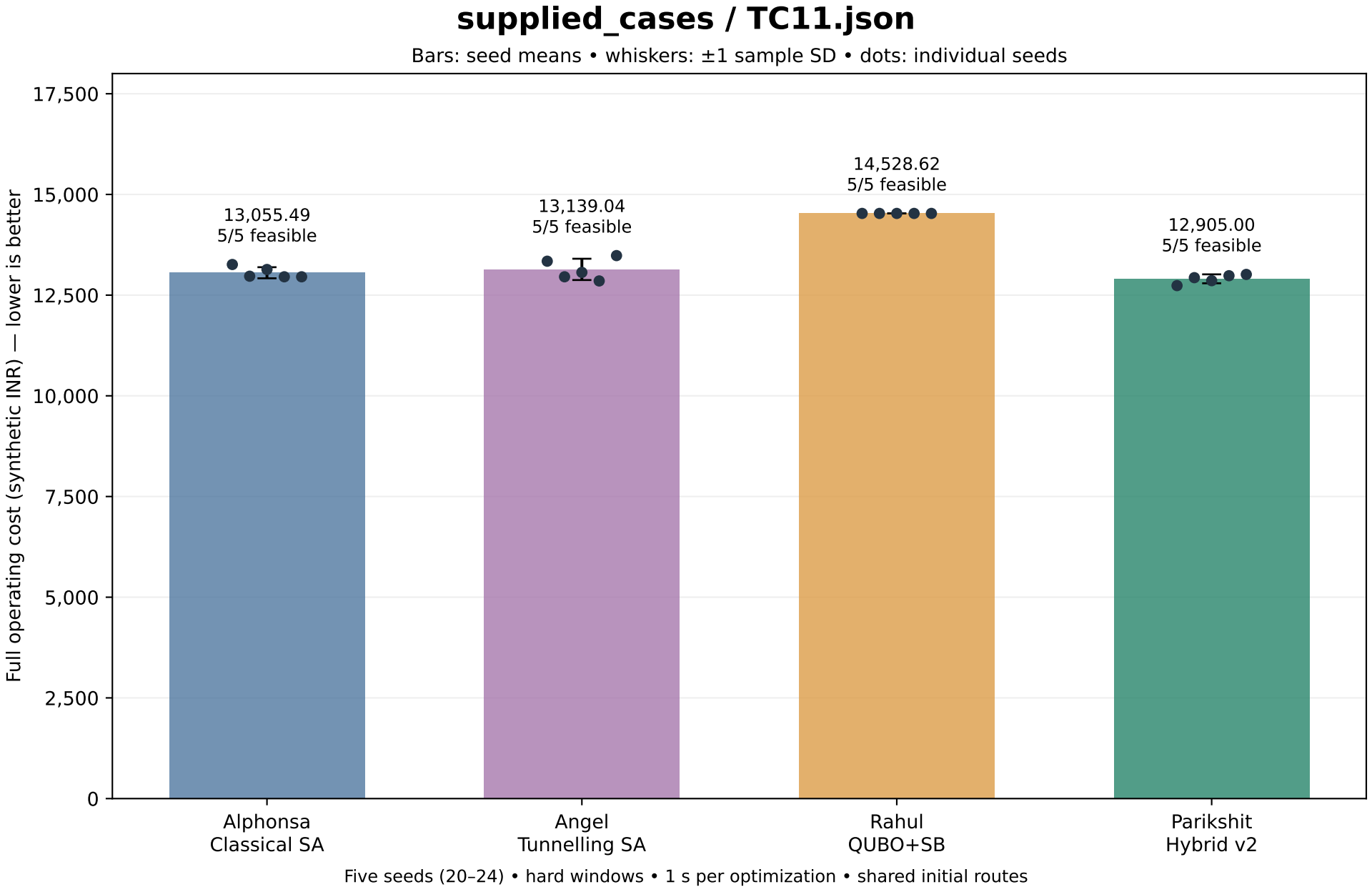

supplied_cases / TC06.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
25,000
20,000
15,000
10,000
5,000
0
21,893.75
5/5 feasible
21,966.52
5/5 feasible
22,532.59
5/5 feasible
21,777.16
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

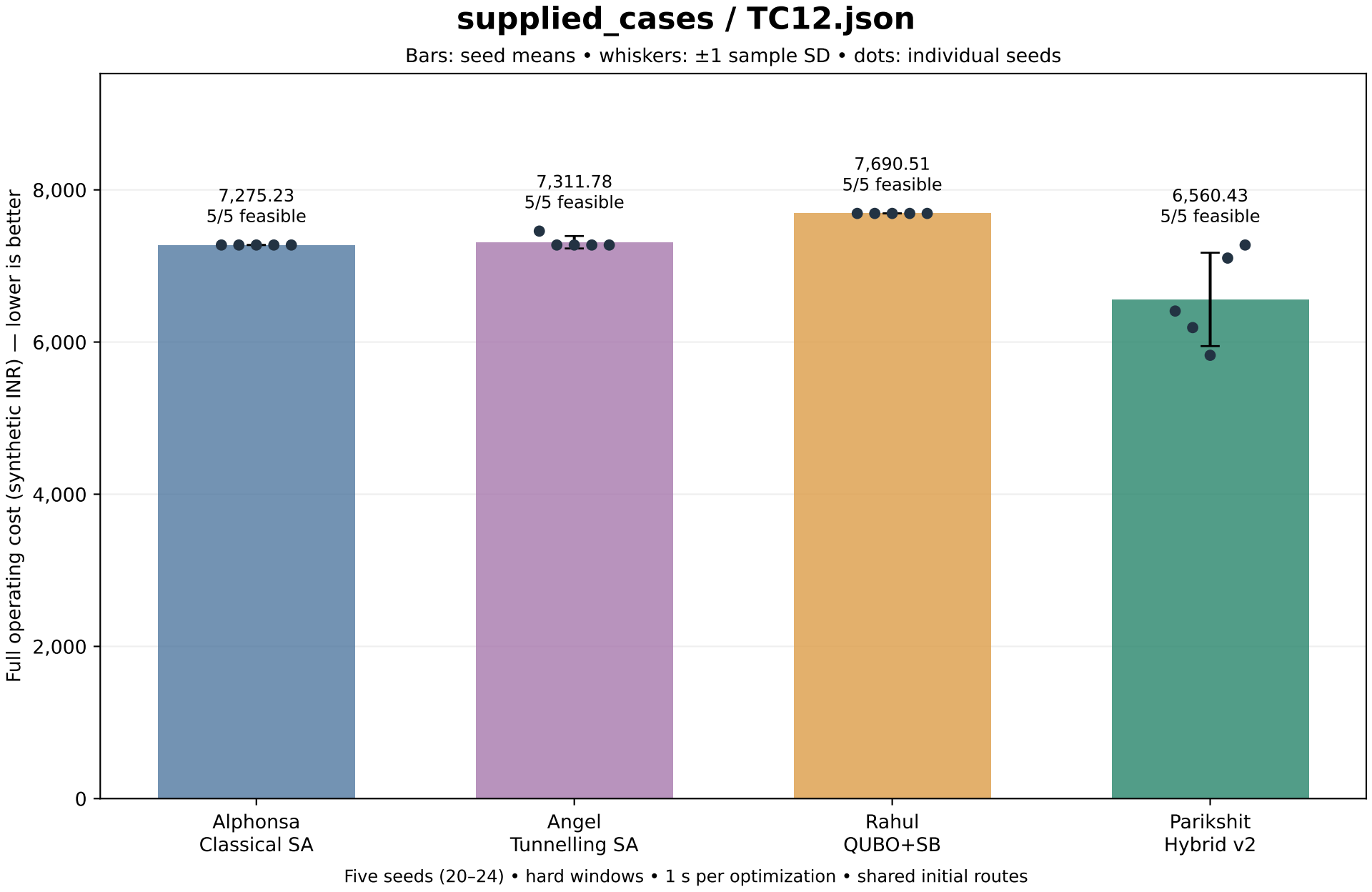

supplied_cases / TC07.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
6,245.51
5/5 feasible
6,245.51
5/5 feasible
6,888.75
5/5 feasible
6,248.01
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

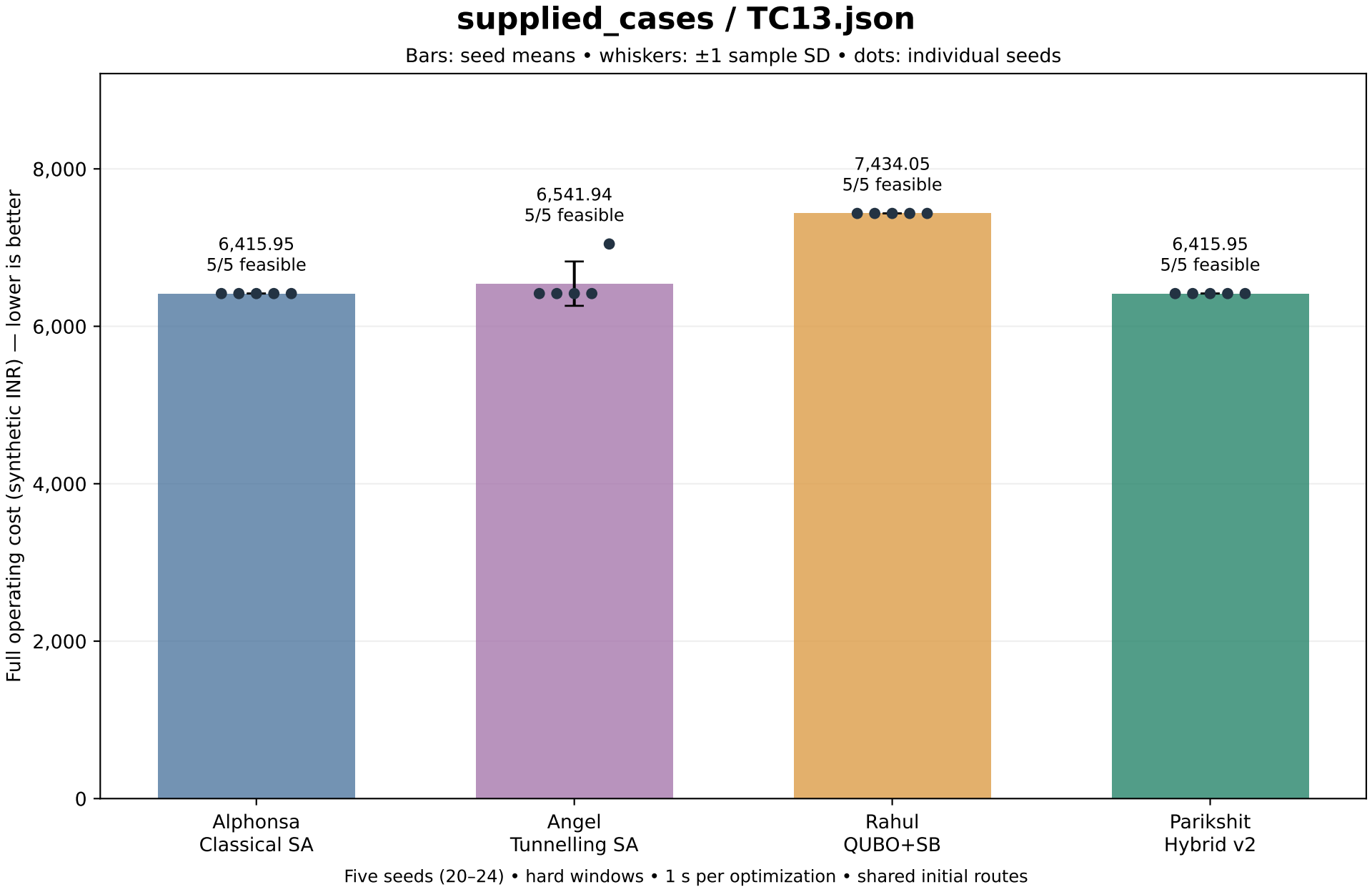

supplied_cases / TC08.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
0
2,000
4,000
6,000
8,000
6,979.05
5/5 feasible
6,875.97
5/5 feasible
7,912.47
5/5 feasible
6,951.94
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC09.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
6,826.41
5/5 feasible
6,828.90
5/5 feasible
7,083.58
5/5 feasible
6,823.92
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC10.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
0
200
400
600
800
1,000
1,200
1,071.35
5/5 feasible
1,071.35
5/5 feasible
1,071.35
5/5 feasible
1,071.35
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC11.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
17,500
15,000
12,500
10,000
7,500
5,000
2,500
0
13,055.49
5/5 feasible
13,139.04
5/5 feasible
14,528.62
5/5 feasible
12,905.00
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC12.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
6,000
4,000
2,000
0
7,275.23
5/5 feasible
7,311.78
5/5 feasible
7,690.51
5/5 feasible
6,560.43
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC13.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
6,000
4,000
2,000
0
6,415.95
5/5 feasible
6,541.94
5/5 feasible
7,434.05
5/5 feasible
6,415.95
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

supplied_cases / TC14.json

INVALID INPUT
Expecting value: line 1 column 1 (char 0)

No finite quality score is available.
This is not a zero-cost result or an infeasibility proof.

Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

supplied_cases / TC15.json

NO FEASIBLE INITIAL SOLUTION
Shared construction failed for all four methods and five seeds.

No finite quality score is available.
This is not a zero-cost result or an infeasibility proof.

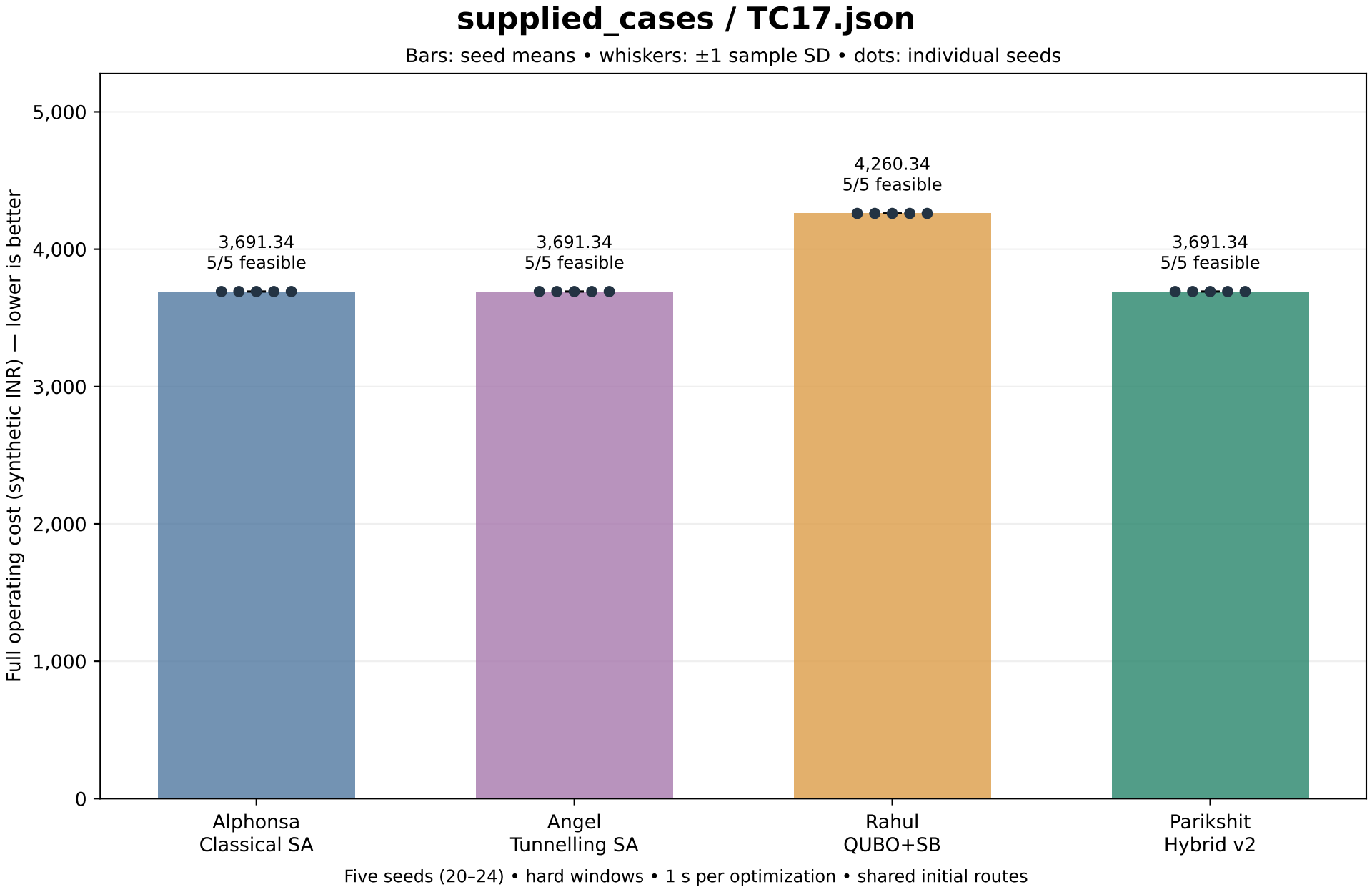
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

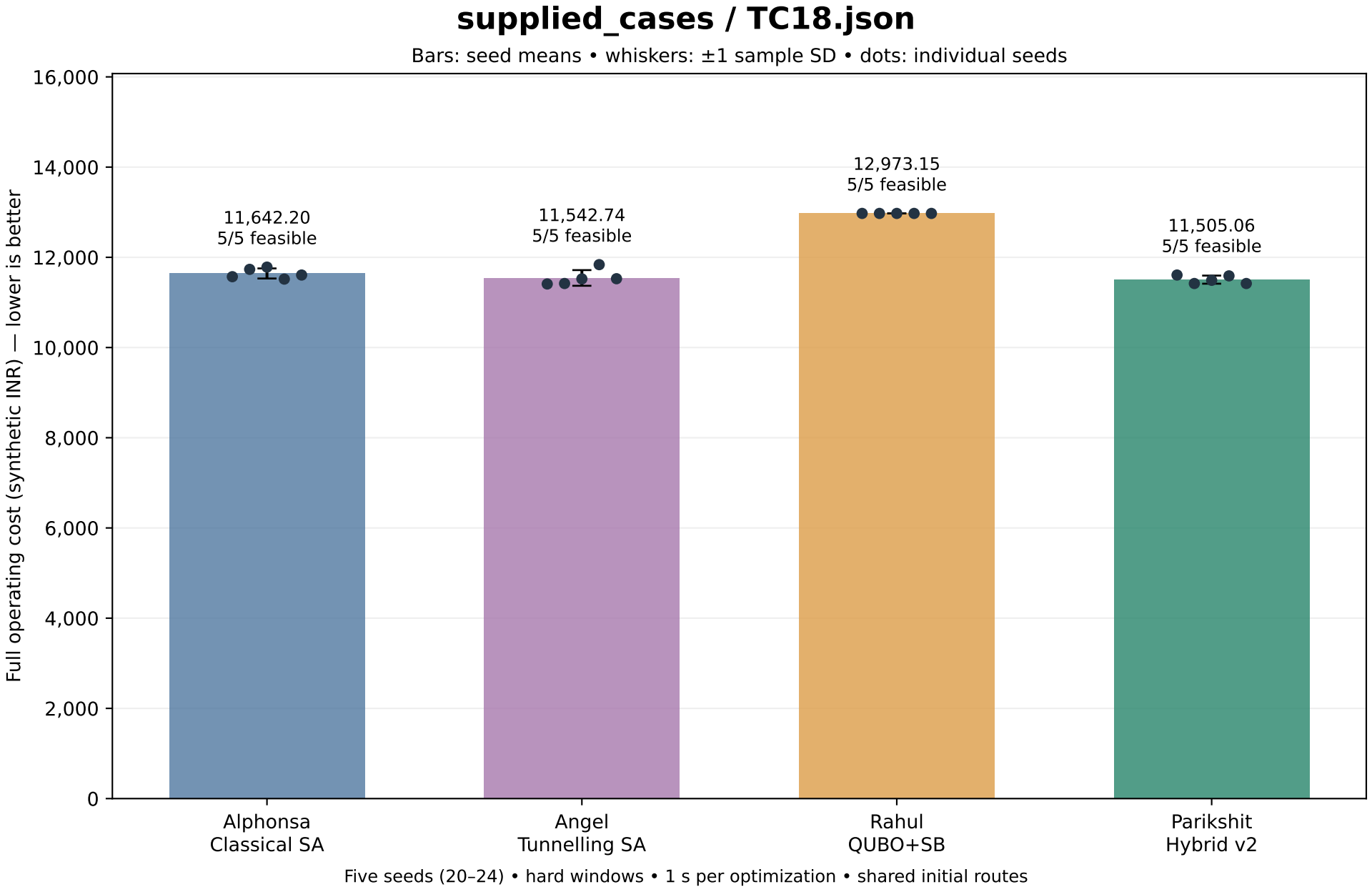
supplied_cases / TC16.json

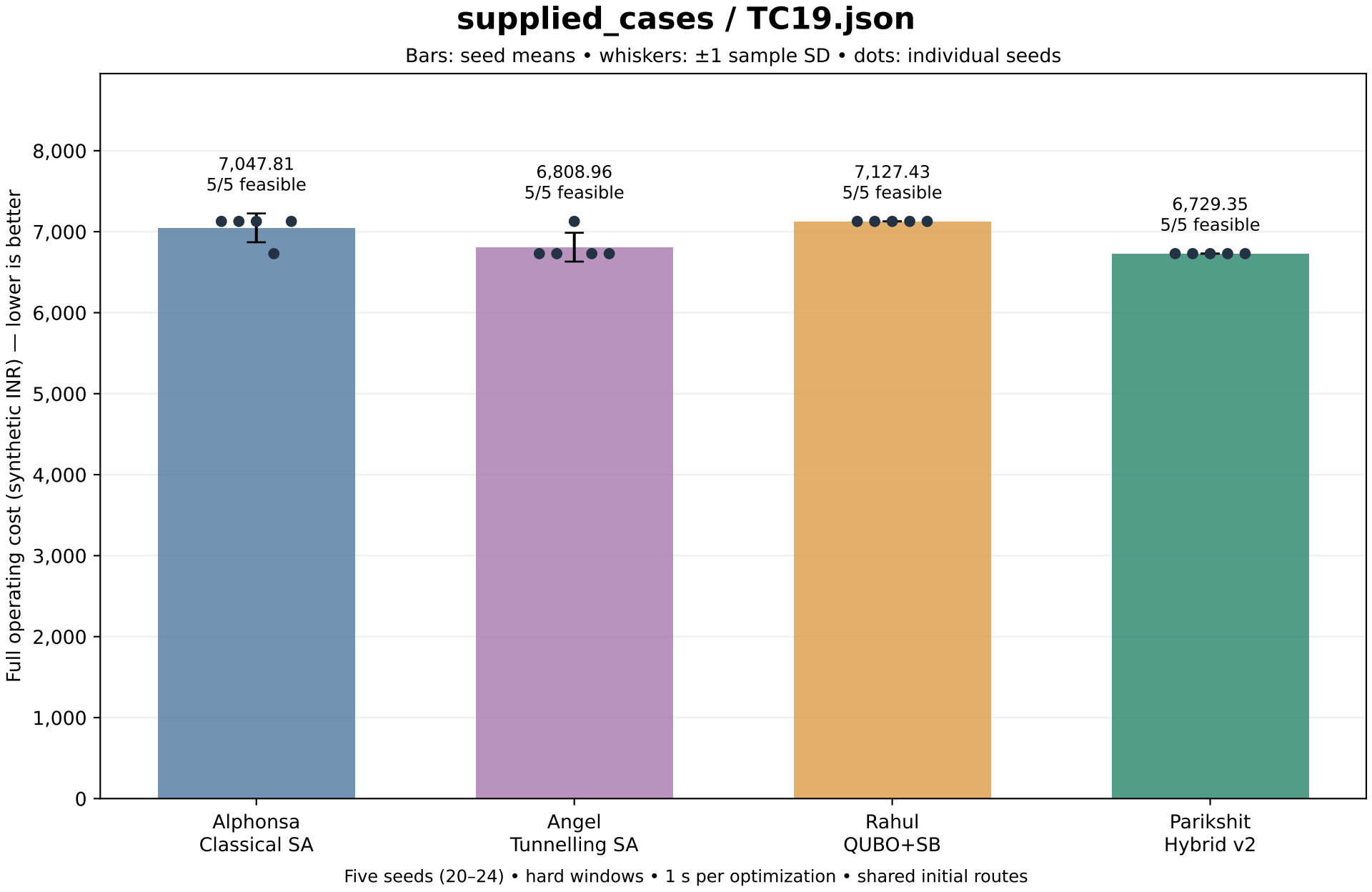
NO FEASIBLE INITIAL SOLUTION
Shared construction failed for all four methods and five seeds.

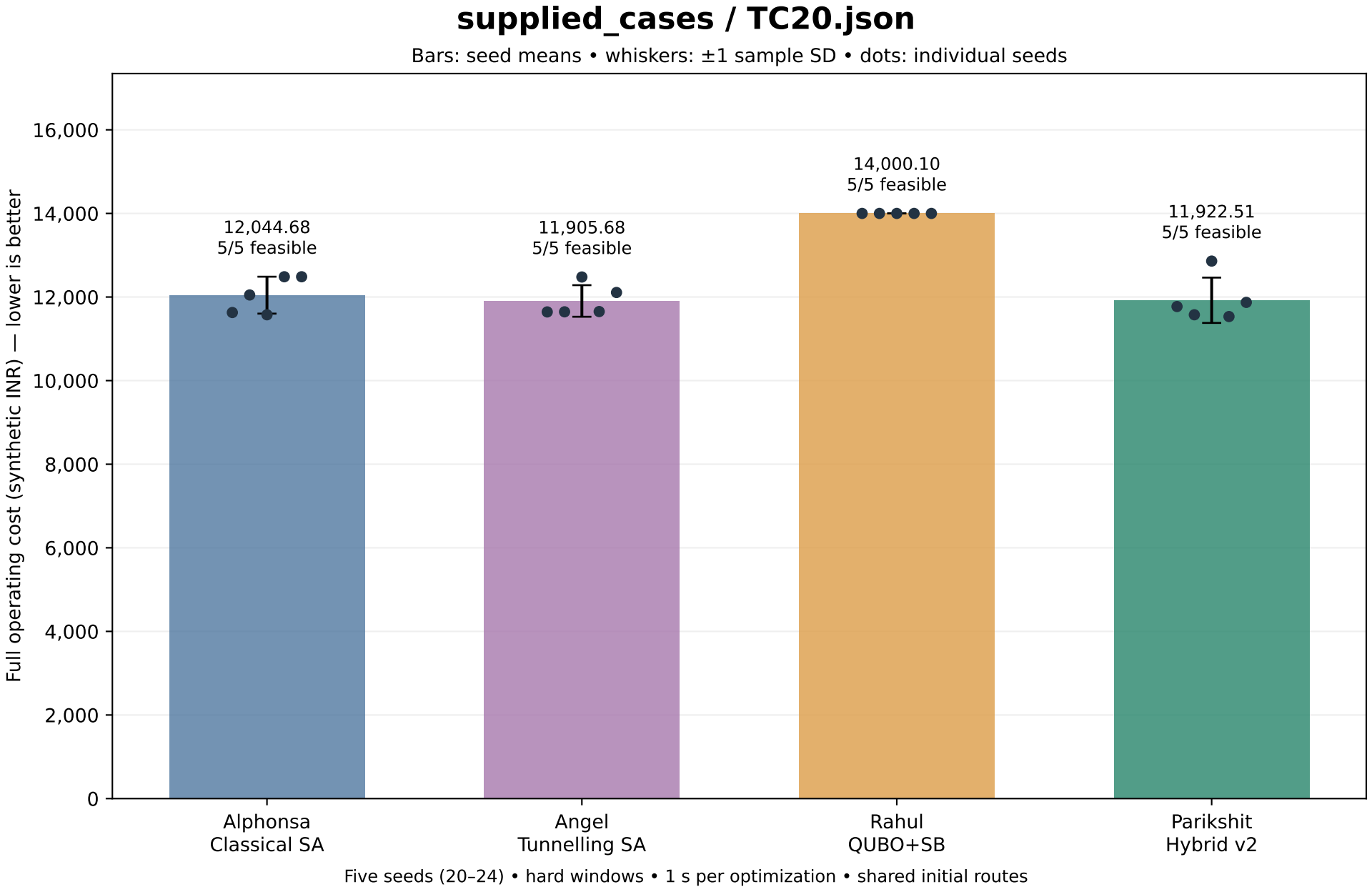
No finite quality score is available.
This is not a zero-cost result or an infeasibility proof.

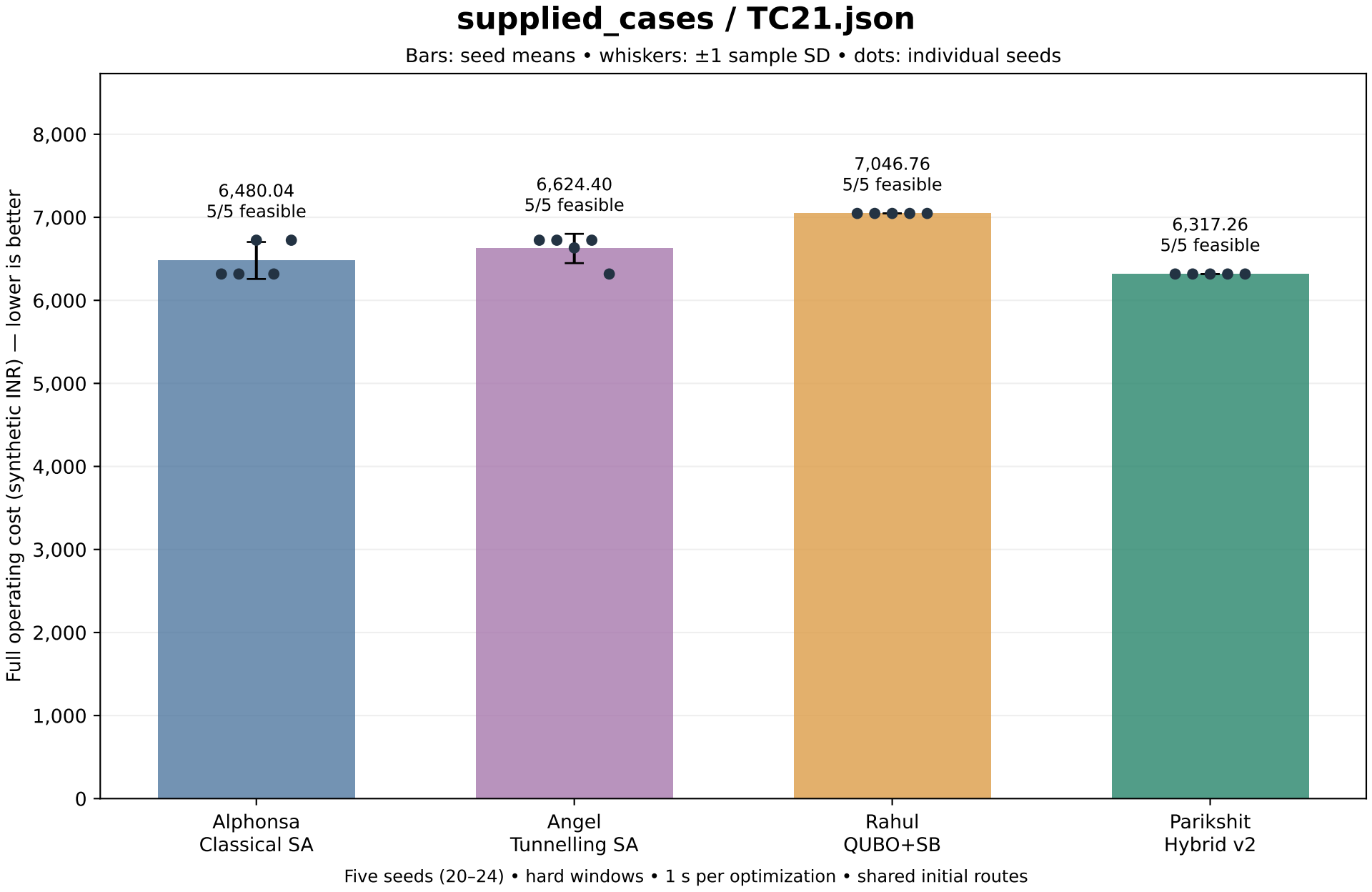
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

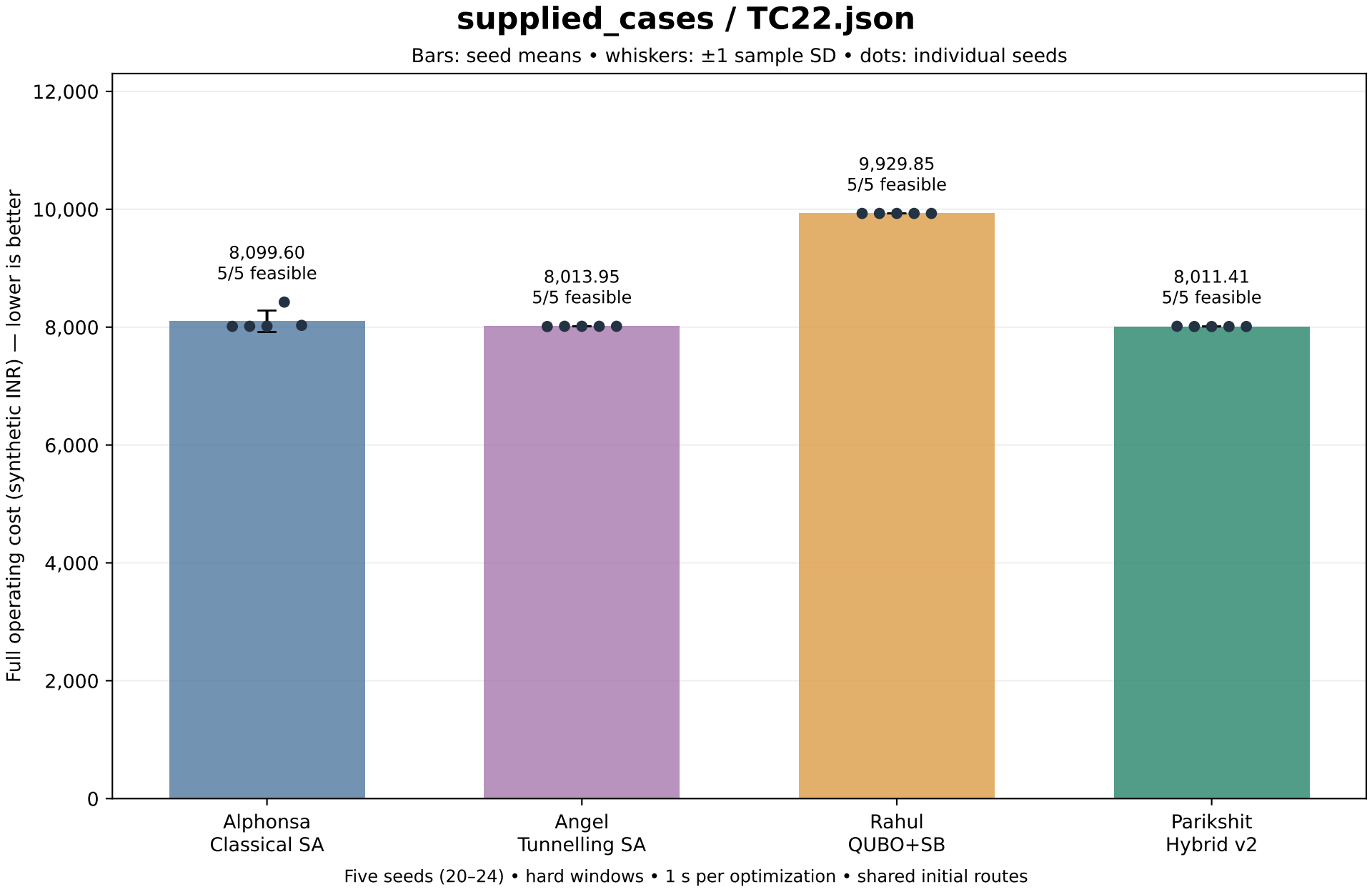

supplied_cases / TC17.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
5,000
4,000
3,000
2,000
1,000
0
3,691.34
5/5 feasible
3,691.34
5/5 feasible
4,260.34
5/5 feasible
3,691.34
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

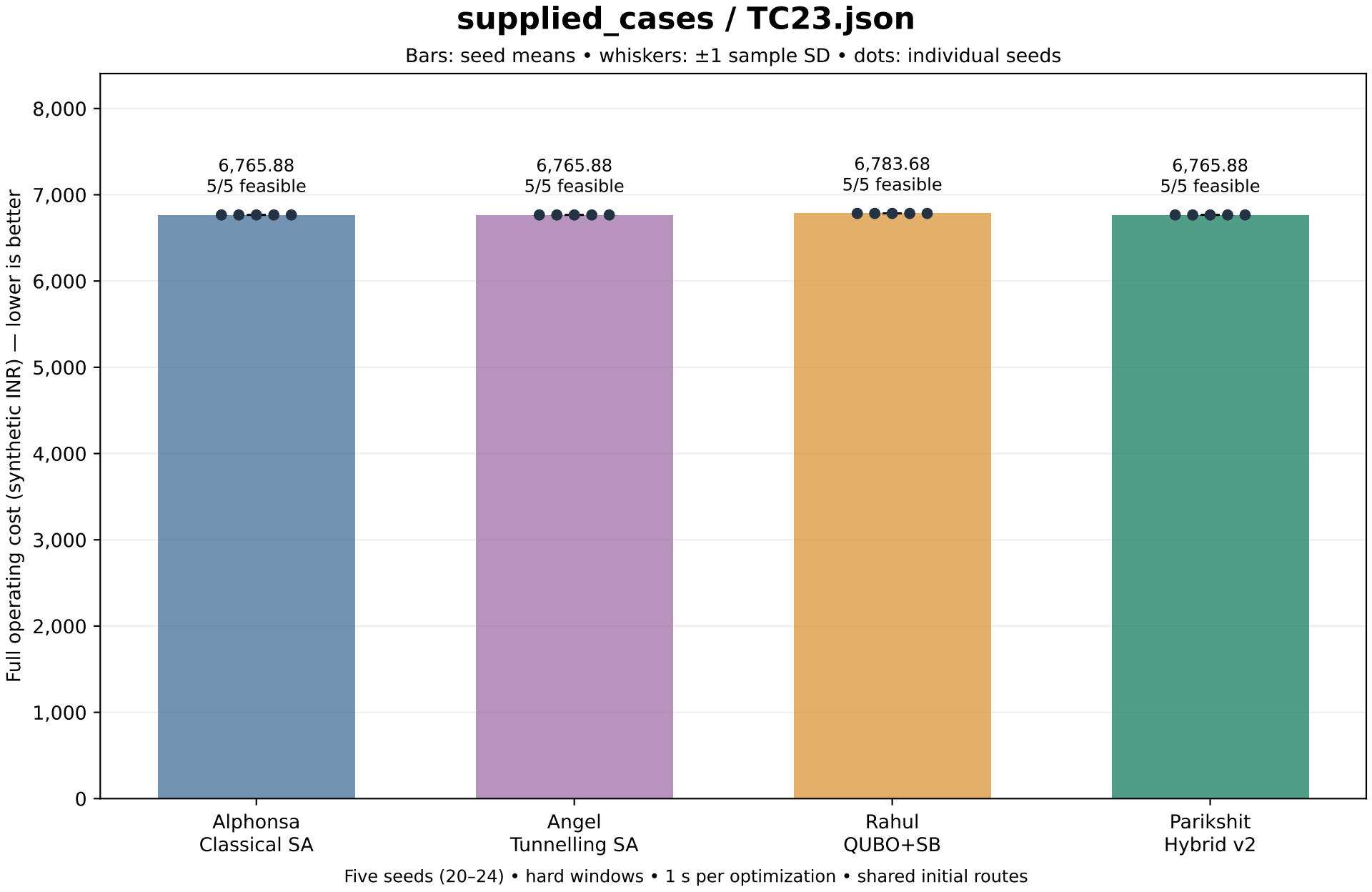

supplied_cases / TC18.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
16,000
14,000
12,000
10,000
8,000
6,000
4,000
2,000
0
11,642.20
5/5 feasible
11,542.74
5/5 feasible
12,973.15
5/5 feasible
11,505.06
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

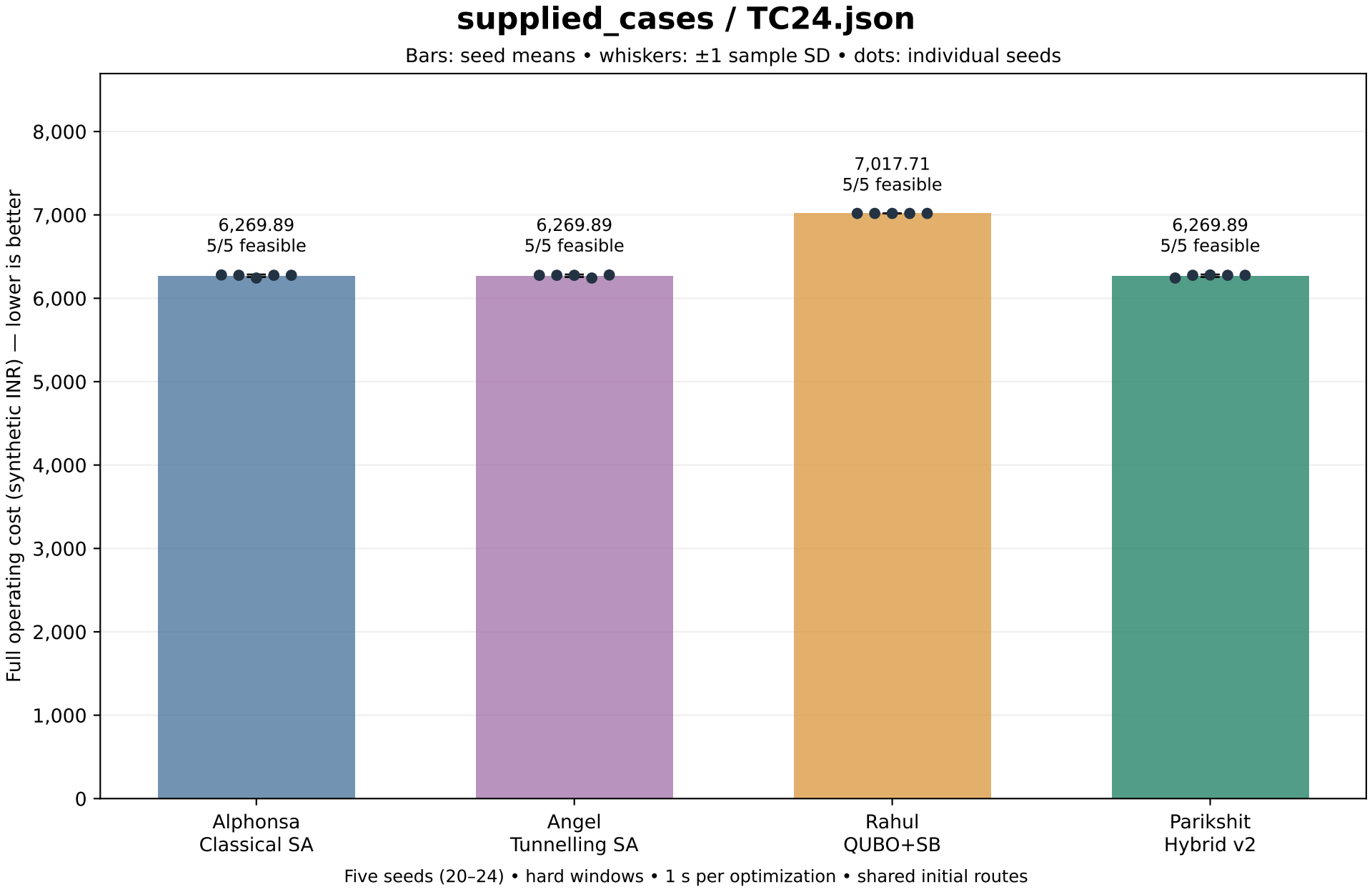

supplied_cases / TC19.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
0
1,000
2,000
3,000
4,000
5,000
6,000
7,000
8,000
7,047.81
5/5 feasible
6,808.96
5/5 feasible
7,127.43
5/5 feasible
6,729.35
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC20.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
16,000
14,000
12,000
10,000
8,000
6,000
4,000
2,000
0
12,044.68
5/5 feasible
11,905.68
5/5 feasible
14,000.10
5/5 feasible
11,922.51
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC21.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
6,480.04
5/5 feasible
6,624.40
5/5 feasible
7,046.76
5/5 feasible
6,317.26
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC22.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
12,000
10,000
8,000
6,000
4,000
2,000
0
8,099.60
5/5 feasible
8,013.95
5/5 feasible
9,929.85
5/5 feasible
8,011.41
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC23.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
6,765.88
5/5 feasible
6,765.88
5/5 feasible
6,783.68
5/5 feasible
6,765.88
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

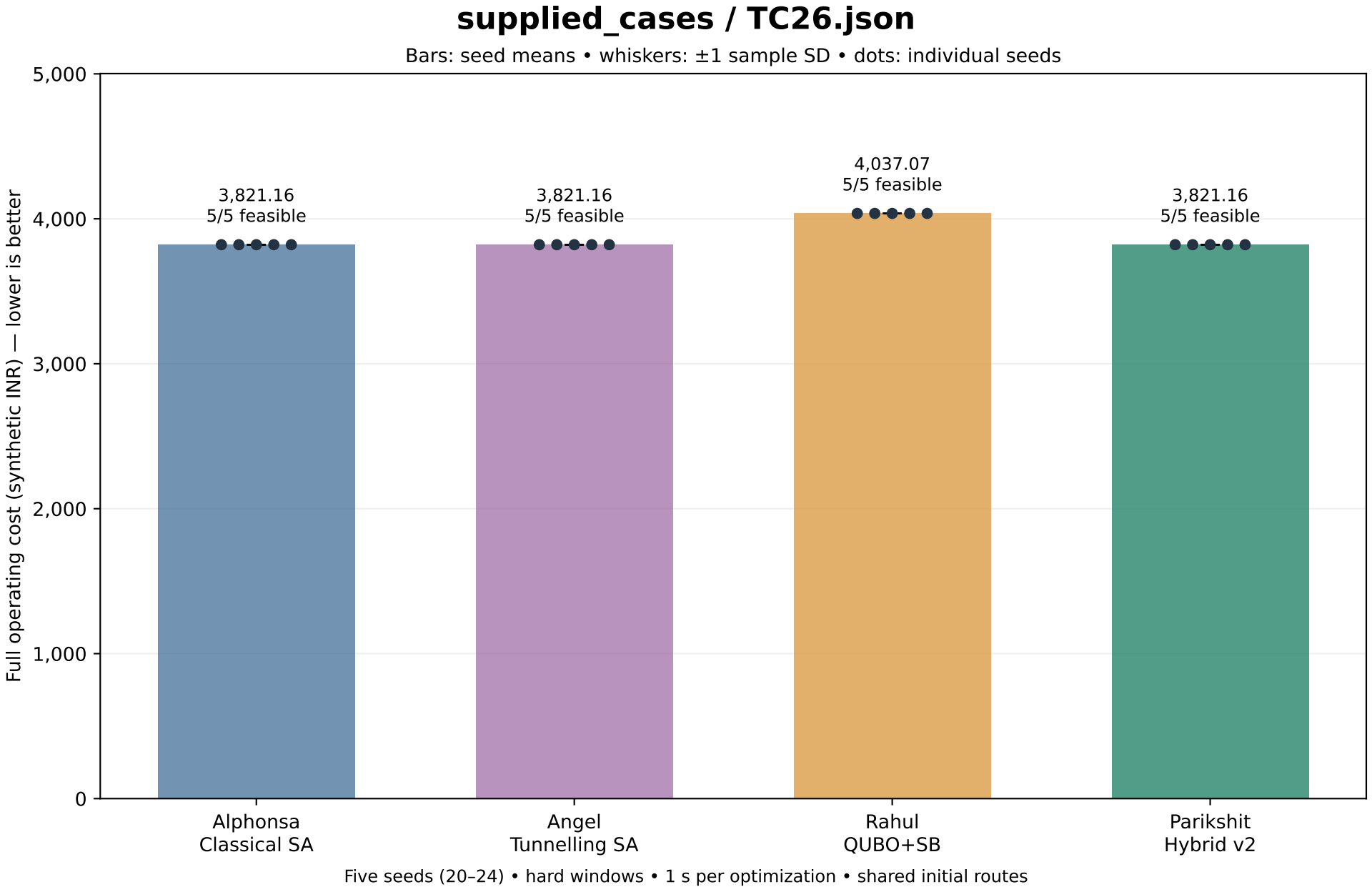

supplied_cases / TC24.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
6,269.89
5/5 feasible
6,269.89
5/5 feasible
7,017.71
5/5 feasible
6,269.89
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

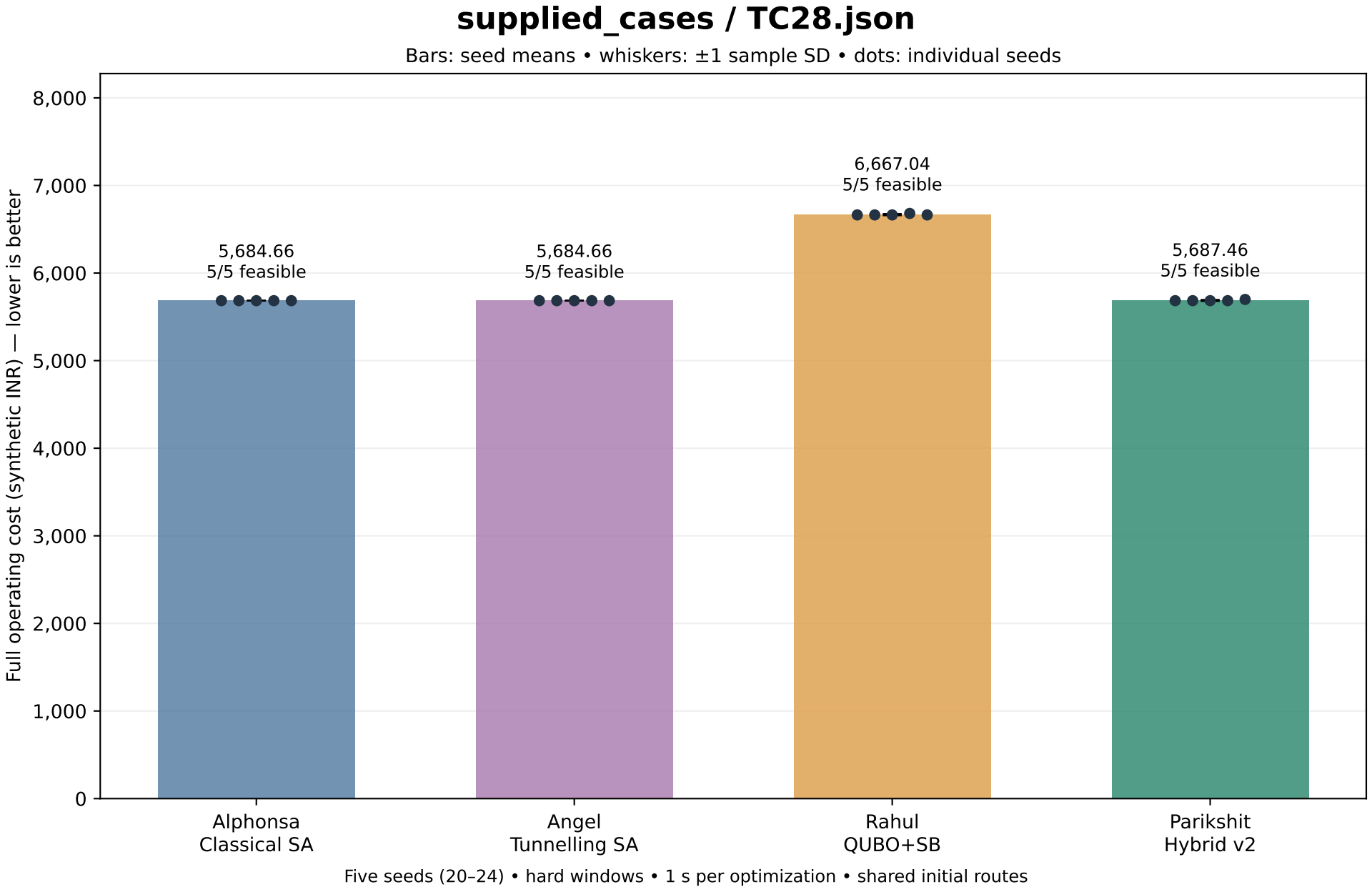
supplied_cases / TC25.json

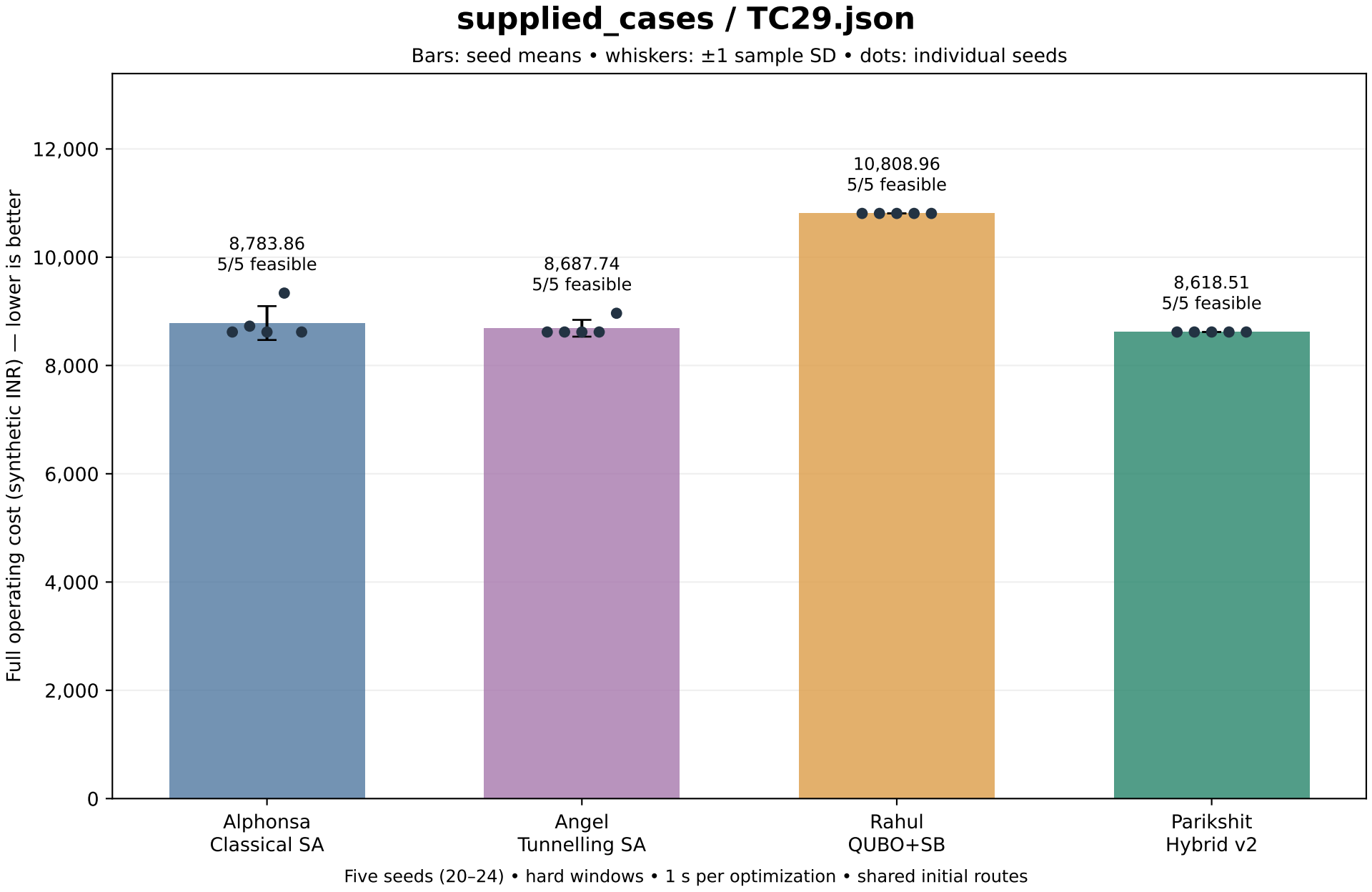
NO FEASIBLE INITIAL SOLUTION
Shared construction failed for all four methods and five seeds.

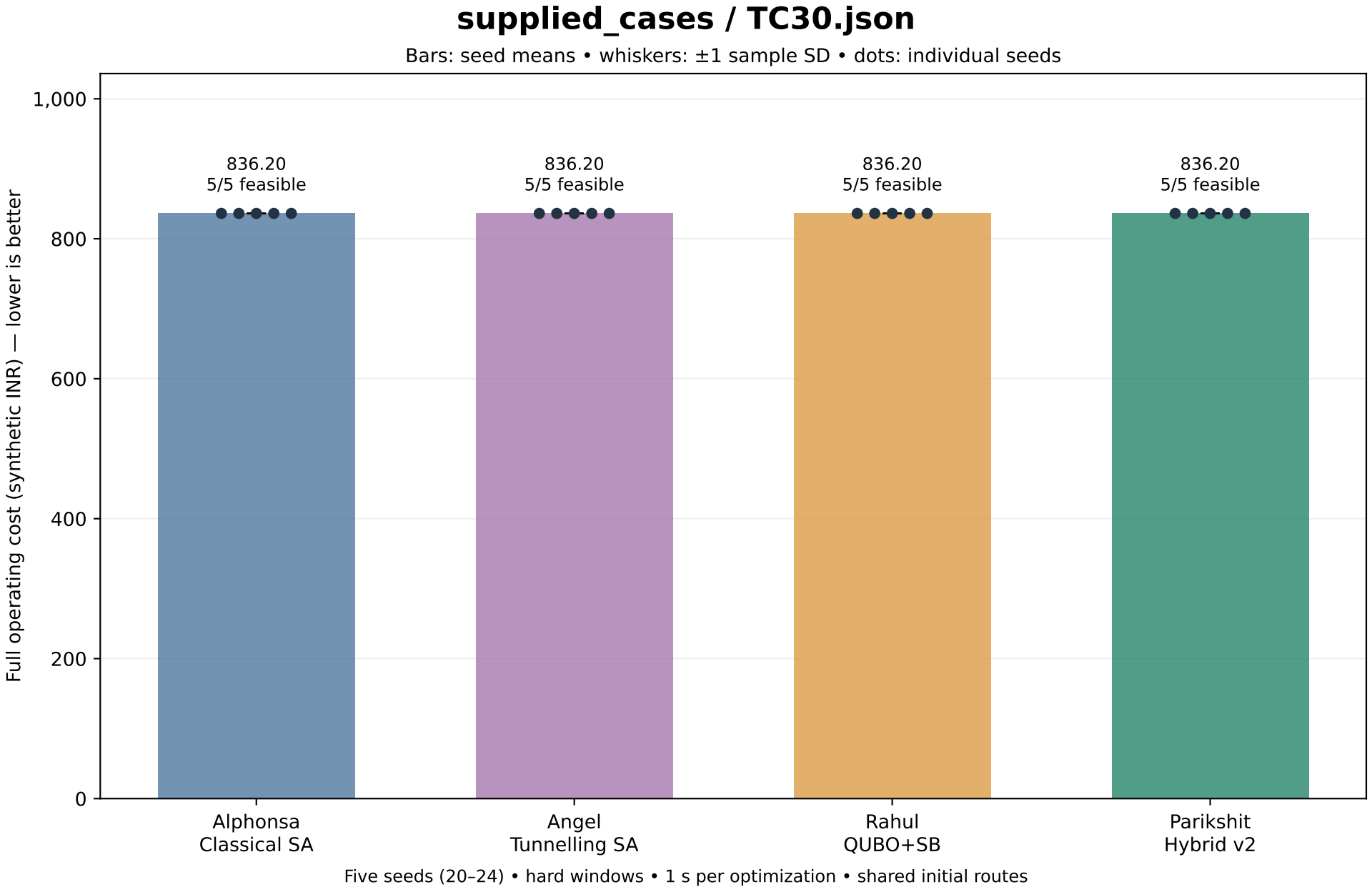
No finite quality score is available.
This is not a zero-cost result or an infeasibility proof.

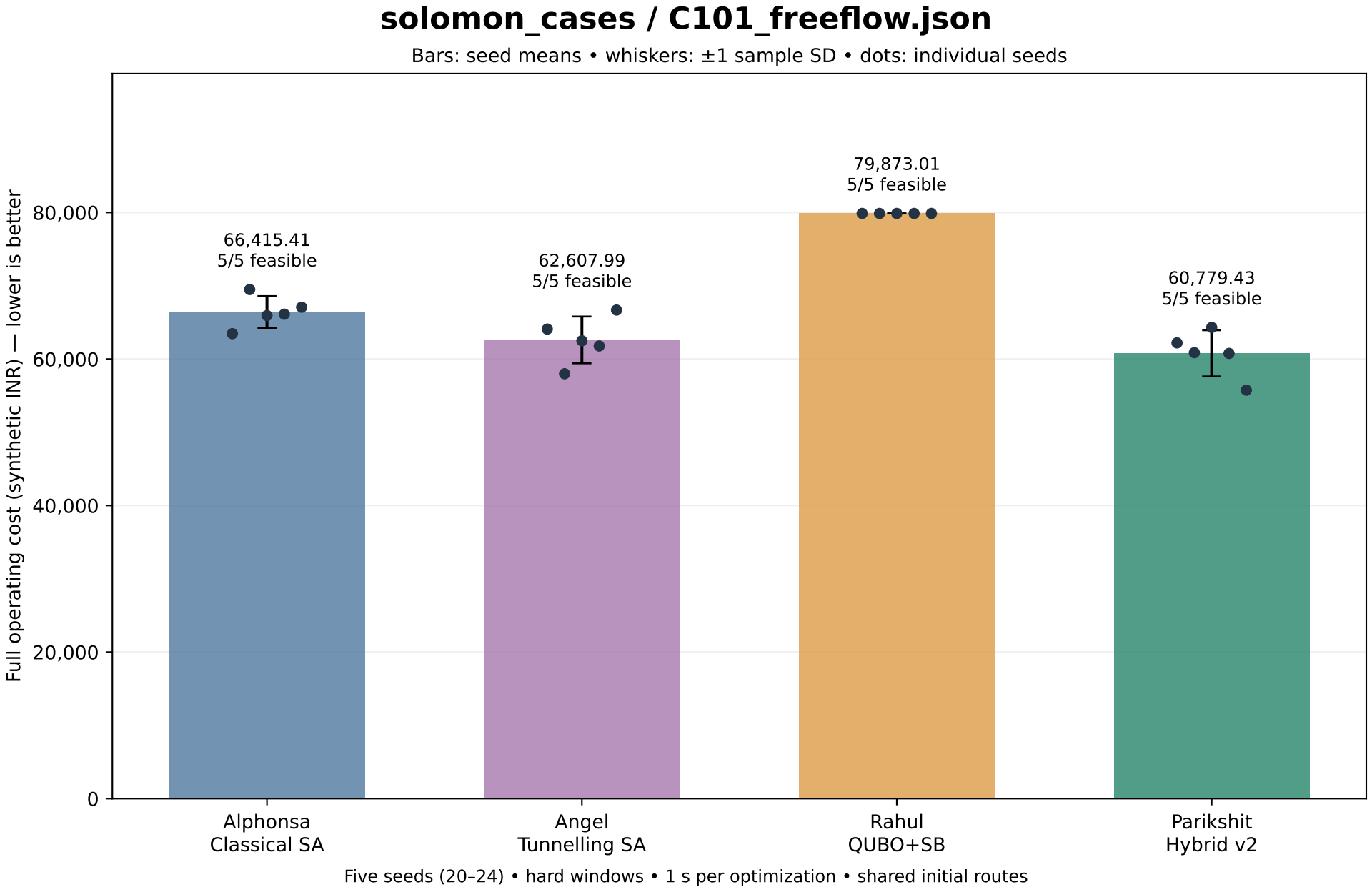
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC26.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
5,000
4,000
3,000
2,000
1,000
0
3,821.16
5/5 feasible
3,821.16
5/5 feasible
4,037.07
5/5 feasible
3,821.16
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

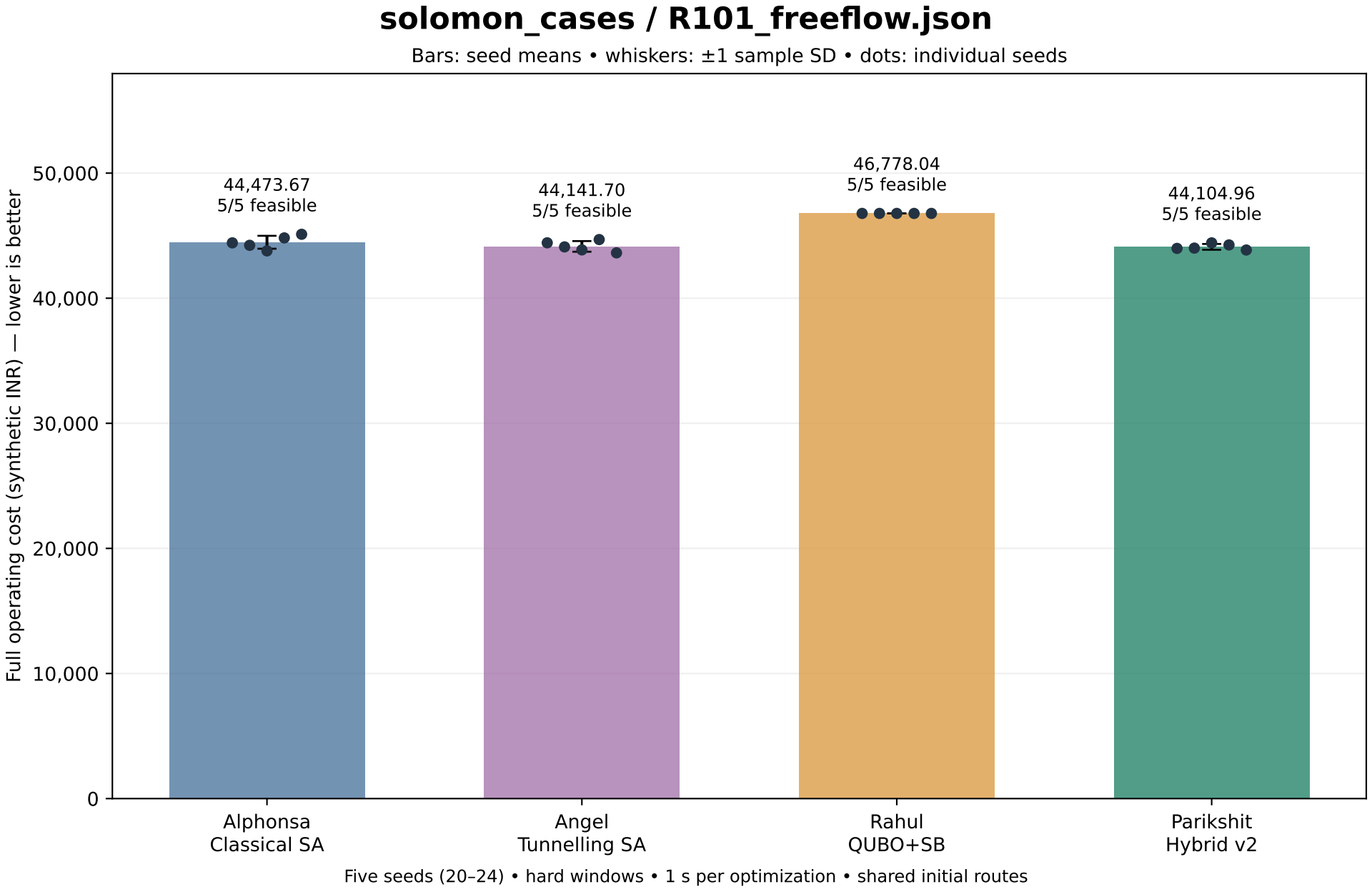

supplied_cases / TC28.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
5,684.66
5/5 feasible
5,684.66
5/5 feasible
6,667.04
5/5 feasible
5,687.46
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC29.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
12,000
10,000
8,000
6,000
4,000
2,000
0
8,783.86
5/5 feasible
8,687.74
5/5 feasible
10,808.96
5/5 feasible
8,618.51
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


supplied_cases / TC30.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
1,000
800
600
400
200
0
836.20
5/5 feasible
836.20
5/5 feasible
836.20
5/5 feasible
836.20
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


solomon_cases / C101_freeflow.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
0
20,000
40,000
60,000
80,000
66,415.41
5/5 feasible
62,607.99
5/5 feasible
79,873.01
5/5 feasible
60,779.43
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


solomon_cases / C101_traffic.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
0
20,000
40,000
60,000
80,000
66,696.89
5/5 feasible
65,114.62
5/5 feasible
79,883.89
5/5 feasible
61,939.11
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

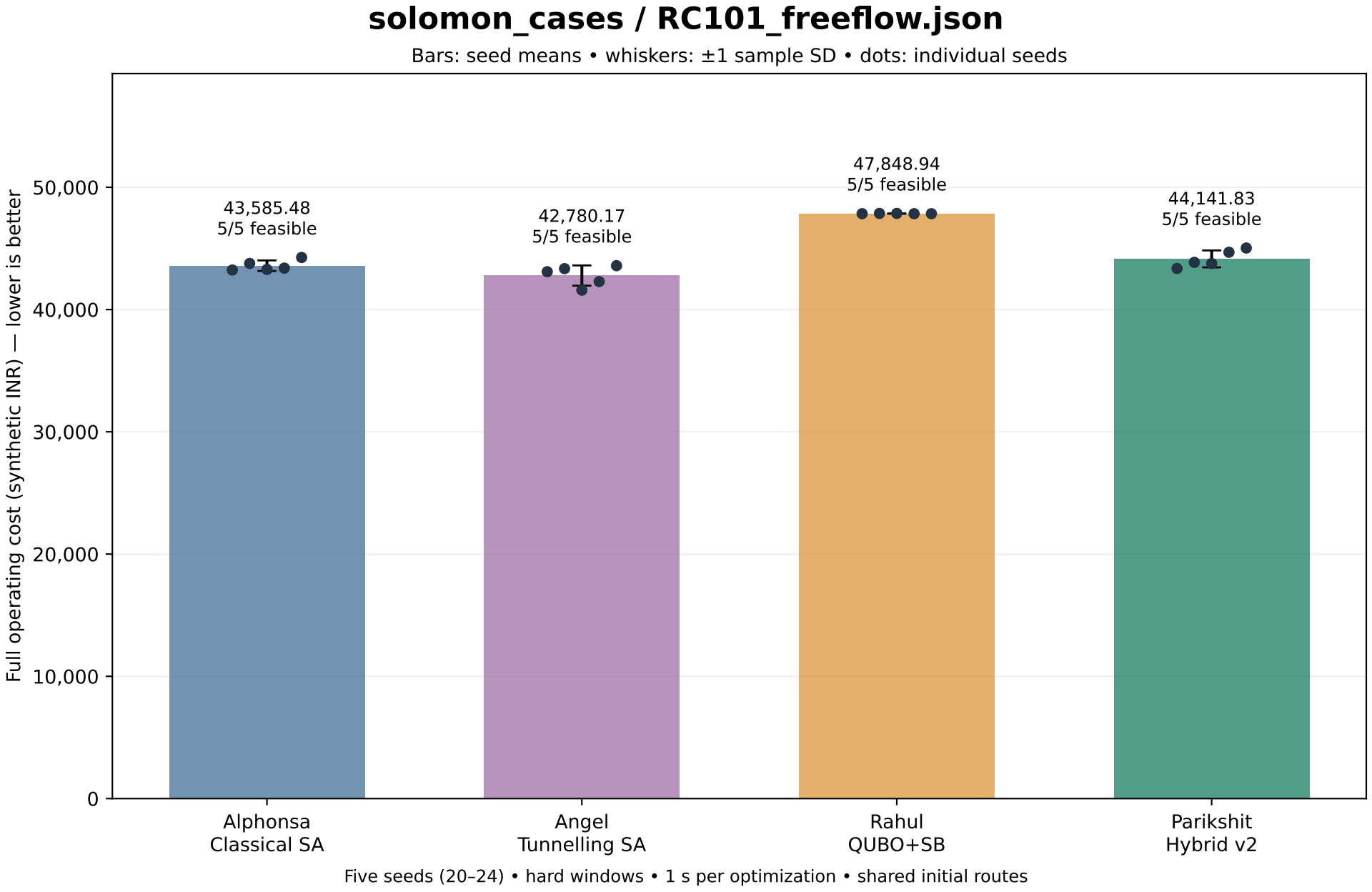

solomon_cases / R101_freeflow.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
50,000
40,000
30,000
20,000
10,000
0
44,473.67
5/5 feasible
44,141.70
5/5 feasible
46,778.04
5/5 feasible
44,104.96
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes

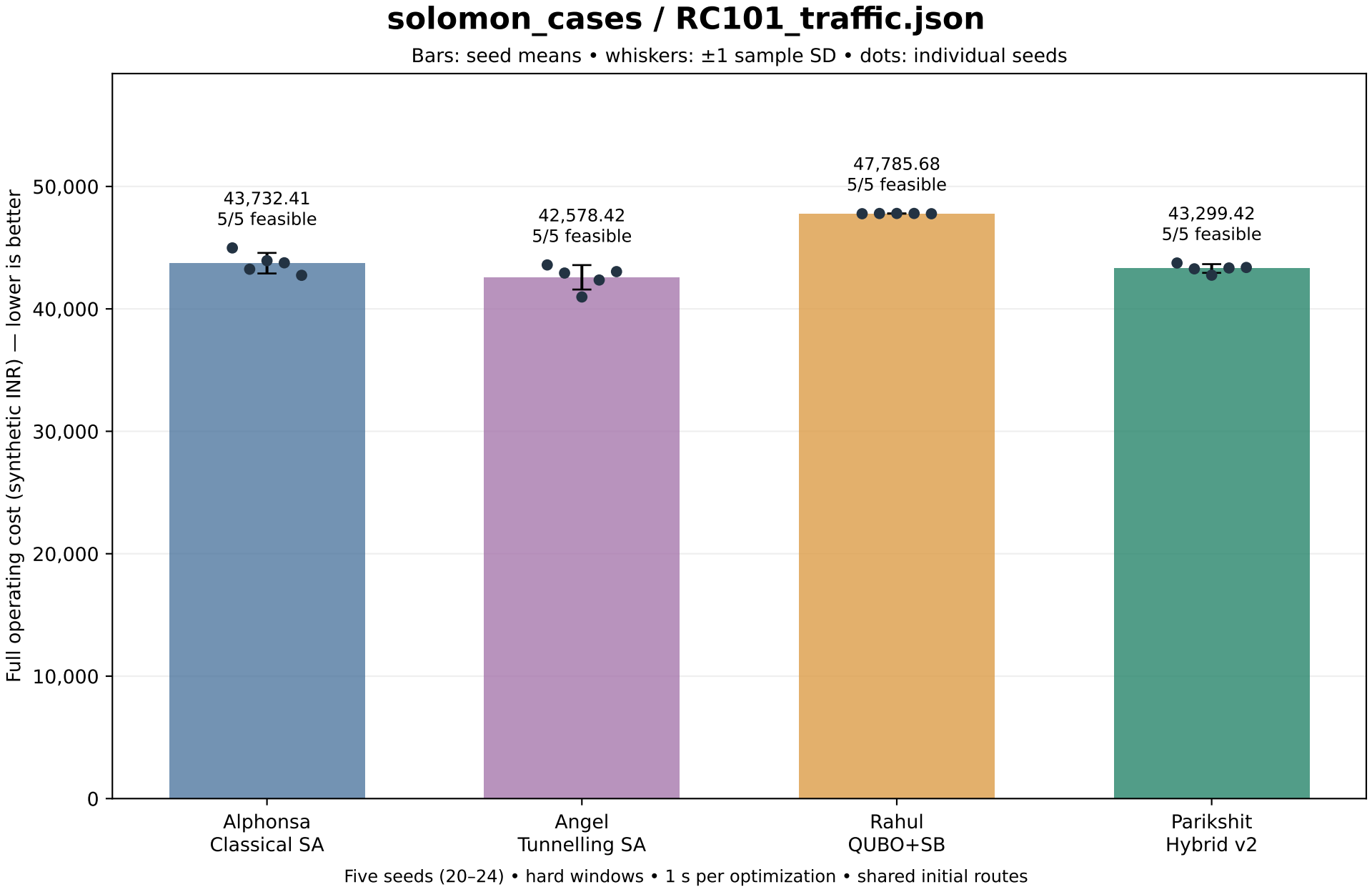
solomon_cases / R101_traffic.json

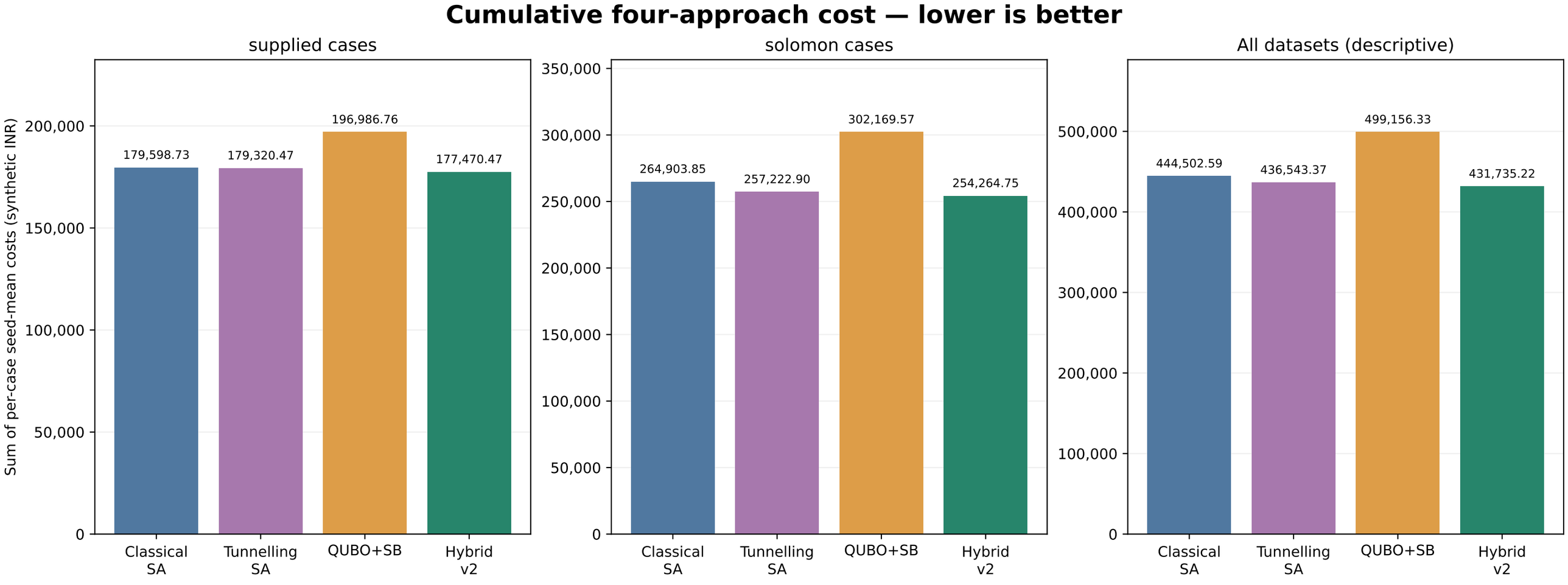
NO FEASIBLE INITIAL SOLUTION
Shared construction failed for all four methods and five seeds.

No finite quality score is available.
This is not a zero-cost result or an infeasibility proof.

Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


solomon_cases / RC101_freeflow.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
50,000
40,000
30,000
20,000
10,000
0
43,585.48
5/5 feasible
42,780.17
5/5 feasible
47,848.94
5/5 feasible
44,141.83
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


solomon_cases / RC101_traffic.json
Bars: seed means • whiskers: ±1 sample SD • dots: individual seeds
Full operating cost (synthetic INR) — lower is better
0
10,000
20,000
30,000
40,000
50,000
43,732.41
5/5 feasible
42,578.42
5/5 feasible
47,785.68
5/5 feasible
43,299.42
5/5 feasible
Alphonsa
Classical SA
Angel
Tunnelling SA
Rahul
QUBO+SB
Parikshit
Hybrid v2
Five seeds (20–24) • hard windows • 1 s per optimization • shared initial routes


Cumulative four-approach cost — lower is better
supplied cases
solomon cases
All datasets (descriptive)
Sum of per-case seed-mean costs (synthetic INR)
0
50,000
100,000
150,000
200,000
250,000
300,000
350,000
400,000
500,000
179,598.73
179,320.47
196,986.76
177,470.47
264,903.85
257,222.90
302,169.57
254,264.75
444,502.59
436,543.37
499,156.33
431,735.22
Classical SA
Tunnelling SA
QUBO+SB
Hybrid v2
30 jointly feasible dataset/case groups; failed/invalid cases have no cost.
Descriptive case basket: expensive cases dominate. Dataset-level statistical inference remains separate.